

DECLARATION

I, **EBENDA ASSENE MARTHE DAVIDA**, hereby declare that this project entitled **“DESIGN AND IMPLEMENTATION OF A DESTINATION VISUALIZER AND VERIFICATION SYSTEM FOR CAMEROON CUSTOMS (DouaneNav)”** is my original work and was carried out by me in partial fulfilment of the requirements for the award of the Bachelor's Degree in Software Engineering.

All ideas, analyses, designs, implementation, testing, and documentation presented in this report are my own except where the work of other authors has been acknowledged through appropriate references in accordance with academic standards.

I further declare that this work has not been submitted, either wholly or in part, to any other institution for the award of any academic qualification.

NAME: EBENDA ASSENE MARTHE DAVIDA

Signature: _____

Date: _____

CERTIFICATION

This is to certify that the project entitled “ **DESIGN AND IMPLEMENTATION OF A DESTINATION VISUALIZER AND VERIFICATION SYSTEM FOR CAMEROON CUSTOMS (DouaneNav)**” was carried out by EBENDA ASSENE MARTHE DAVIDA under my supervision and was examined and approved for submission to the Department of Software Engineering in partial fulfilment of the requirements for the award of the Bachelor's Degree in Software Engineering.

SUPERVISOR: _____

Signature: _____

Date: _____

DEDICATION

To my Family

ACKNOWLEDGEMENT

My sincere appreciation goes to my academic supervisor, **Engr. ABANJI DIEUDONNE**, for his invaluable guidance, constructive advice, and continuous support throughout this project.

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ABSTRACT

Cameroonian customs officers often consult several documents, including customs declarations, bills of lading, invoices, packing lists, and transport particulars, to verify the declared destination of shipments. The lack of visual destination information makes this process time-consuming and limits field officers' advance visibility. This study proposes **DouaneNav**, a mobile-first decision support subsystem designed to complement CAMCIS. The system enables customs officers to retrieve declaration information, search by declaration number or truck plate, visualise declared destinations using OpenStreetMap, and record inspection outcomes. A web dashboard supports administrative monitoring and synchronisation of inspection data. The system was developed using React Native, React.js, Node.js, Express, MongoDB, and JWT-based role-based access control. AI-based document analysis is reserved for future development. The study demonstrates the feasibility of using mobile and map-based technologies to improve destination verification and inspection support in Cameroon's customs operations.

Keywords: Customs Digitisation, CAMCIS, Destination Visualisation, Mobile Decision Support, Inspection Support, OpenStreetMap, Cameroon.

RESUME

La vérification de la destination déclarée d'une marchandise constitue un défi pour les agents des douanes camerounaises, car les informations nécessaires sont réparties dans plusieurs documents, notamment la déclaration CAMCIS, la facture, le connaissance, la liste de colisage et les informations du transporteur. Cette situation entraîne des vérifications manuelles et limite la visibilité des agents lors des contrôles sur le terrain.

Ce projet présente **DouaneNav**, un sous-système mobile d'aide à la décision conçu pour compléter CAMCIS sans le remplacer. L'application permet aux agents de rechercher les déclarations, de consulter les informations relatives aux marchandises, de visualiser les destinations déclarées sur une carte et d'enregistrer les résultats des inspections. Un tableau de bord web permet également aux administrateurs et superviseurs de gérer les utilisateurs, de suivre les synchronisations et de consulter les statistiques.

Les résultats montrent que DouaneNav peut faciliter la vérification des destinations, la préparation des inspections et la gestion des rapports de terrain. L'analyse documentaire par intelligence artificielle est prévue comme une amélioration future.

Mots-clés : Douanes camerounaises, CAMCIS, Visualisation des destinations, Application mobile, Inspection, Synchronisation, Aide à la décision.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASYCUDA	Automated System for Customs Data
CAMCIS	Cameroon Customs Information System
CEPI	Cameroon Economic Policy Institute
DGD	Direction Générale des Douanes
DSS	Decision Support System
GIS	Geographic Information Systems
GPS	Global Positioning System
HCI	Human-Computer Interaction
HS	Harmonised System
JWT	JSON Web Token
NLP	Natural Language Processing
OCR	Optical Character Recognition
PAGODE	Procédures Automatisées de Gestion des Opérations de la Douane et du commerce Extérieur
RBAC	Role-Based Access Control
SYDONIA	Système Douanier Automatisé
UNCTAD	United Nations Conference on Trade and Development

CHAPTER ONE

INTRODUCTION

This chapter introduces the study of DouaneNav, a mobile-first decision support subsystem for visualising declared destinations and supporting field inspections within the Cameroonian customs environment. It presents the historical, conceptual, theoretical, and contextual background of the study, followed by the problem statement, objectives, research questions, significance, justification, delimitation, and organisation of the study.

1.1 Background of the Study

1.1.1 Historical Background

Customs control has, for centuries, depended on paper documentation as proof of the origin, nature, and destination of traded goods. Cameroon Customs first computerised its declaration process in 1984 with PAGODE (Douanes; Douanes) (Procédures Automatisées de Gestion des Opérations de la Douane et du commerce Extérieur), a system that operated principally in the Littoral customs sector around the port of Douala and, at its peak, processed the large majority of the country's customs revenue. Over roughly two decades of use, PAGODE became increasingly difficult to maintain and update, and by the early 2000s it no longer met the technical requirements of a modernising customs administration.

Following pilot deployments between 2004 and 2006, Cameroon Customs officially replaced PAGODE with SYDONIA, the French designation for the Automated System for Customs Data (ASYCUDA, or Système Douanier Automatisé), a customs declaration-processing system owned and maintained by the United Nations Conference on Trade and Development (UNCTAD), the United Nations body responsible for trade and development policy. SYDONIA (specifically its SYDONIA++/ASYCUDA++ version) became Cameroon's core declaration-processing platform from 1 January 2007. While SYDONIA represented a substantial improvement over PAGODE, automating the full declaration lifecycle for the first time, it had originally been designed for statistical purposes and, despite later enhancements, showed limitations in fully supporting clearance procedures as trade volumes grew (Direction Générale des Douanes, n.d.).

On 5 June 2020, Cameroon's Minister of Finance officially launched CAMCIS as the successor to SYDONIA. CAMCIS is a modern, web-based, client-server platform developed under a

public-private partnership signed in September 2015 between the Government of Cameroon and CAMPASS, a company backed by Korean capital and technical expertise, with the system modelled on the UNI-PASS platform used by the Korea Customs Service. CAMCIS was introduced to accelerate clearance, reduce face-to-face engagement and associated opportunities for fraud, and allow remote, browser-based access to customs procedures.

Although CAMCIS automates the processing of customs declarations, this study explores the development of a complementary, mobile-first subsystem dedicated to destination visualisation and field inspection support, in order to further assist customs officers operating away from the point of declaration. The more recent emergence of accessible mapping platforms and mobile development frameworks has made it practical to build such a subsystem at reasonable cost, with AI-assisted document analysis identified as a natural, and architecturally accommodated, future extension.

1.1.2 Conceptual Background

1.1.2.1 Customs Declaration

A customs declaration is a formal statement submitted to a customs authority containing information about goods being imported, exported, or transported in transit. It provides customs authorities with the information required to identify the goods, determine their customs treatment, assess applicable duties and taxes, and perform the necessary control procedures. A customs declaration generally contains information such as the importer or exporter, description of the goods, quantity, value, country of origin, means of transport, and declared destination.

In Cameroon, customs declarations are processed through the Cameroon Customs Information System (CAMCIS), which serves as the principal digital platform for customs declaration and clearance procedures. The declaration therefore constitutes an important source of information for customs officers because it provides the official information against which the movement and inspection of goods can be assessed.

Within DouaneNav, the customs declaration serves as the primary source of shipment information retrieved for field operations. Relevant information, including the declaration number, importer, transport company, truck and driver particulars, declared destination, and declaration status, is presented to authorised officers through the mobile application. DouaneNav does not modify the original customs declaration; instead, it uses the relevant

information contained in the declaration to support destination visualisation, inspection preparation, and the recording of inspection results.

1.1.2.2 Cameroon Customs Information System (CAMCIS)

The Cameroon Customs Information System (CAMCIS) is the digital platform used by the Cameroonian Customs Administration to manage and process customs operations. It was introduced as part of the modernization of customs procedures and replaced the previous SYDONIA system. CAMCIS provides a digital environment for the processing of customs declarations and other related customs procedures, reducing dependence on manual and paper-based processes and facilitating access to customs information.

CAMCIS plays a central role in the present study because DouaneNav is designed as a complementary subsystem rather than as a replacement for the existing customs platform. The proposed system is intended to obtain relevant declaration information from CAMCIS and present selected information in a form that is more suitable for mobile field operations. During development and testing, however, live CAMCIS data is not used because the project does not have authorised access to the production platform. Instead, equivalent declaration data is simulated locally to demonstrate the intended integration and operational workflow.

The relationship between CAMCIS and DouaneNav is therefore based on complementary functions. CAMCIS remains responsible for the management and processing of customs declarations, while DouaneNav focuses on destination visualisation, mobile access to relevant declaration information, field inspection support, and the preparation and synchronisation of inspection reports. This approach allows the proposed system to extend the operational usefulness of existing customs infrastructure without attempting to duplicate or replace the functions of CAMCIS.

1.1.2.3 Decision Support System (DSS)

A Decision Support System (DSS) is a computer-based information system designed to assist users in making decisions by providing relevant information, analysis, and tools for evaluating available alternatives. Unlike systems that completely automate decision-making, a DSS generally supports the human decision-maker by organising information and presenting it in a form that facilitates interpretation and judgement. The final decision remains with the user,

while the system reduces the time and effort required to obtain and process relevant information.

Decision Support Systems are particularly useful in situations where decisions involve a combination of available data, operational procedures, and human judgement. In such environments, a system can retrieve relevant information, organise it according to the user's needs, and present it in a way that helps the user assess a situation and determine an appropriate course of action.

In the context of this study, DouaneNav functions as a mobile-first decision support subsystem for customs field operations. The system does not independently decide whether a shipment should be inspected or determine the outcome of an inspection. Instead, it provides Customs Officers with relevant declaration information, visualises the declared destination on an interactive map, and provides tools for recording inspection findings. By reducing the effort required to locate and interpret destination-related information, DouaneNav is intended to support officers in making better-informed operational decisions while preserving human responsibility for inspection and enforcement decisions.

1.1.2.4 Mobile Decision Support

Mobile decision support refers to the use of mobile computing technologies to provide users with relevant information and decision-support functions while they are away from a fixed workstation. Unlike traditional decision support systems that are commonly accessed through desktop computers, mobile decision support systems are designed to provide information and operational tools through smartphones or other portable devices. This allows users to access, process, and record information directly within the environment where decisions and operational activities take place.

Mobile decision support is particularly relevant to field-based activities where users may need to work while moving between locations or under conditions where access to a conventional computer is limited. Such systems can provide rapid access to relevant information, search functions, location-based information, data capture, and reporting capabilities. Their effectiveness depends on presenting essential information clearly while limiting unnecessary interaction, particularly when users are operating under time constraints or variable network conditions.

In DouaneNav, mobile decision support is implemented through the React Native application used by Customs Officers during field operations. The application provides access to relevant declaration information, allows officers to search for consignments, displays declared destinations on an interactive map, and provides an interface for recording inspection outcomes, comments, and photographs. The mobile-first approach therefore places the operational functions required by field officers on a portable device, allowing destination information and inspection tools to remain accessible outside the conventional customs office environment.

1.1.2.5 Destination Visualisation

Destination visualisation refers to the representation of a location or destination through a visual geographic interface, such as an interactive digital map, rather than relying solely on textual descriptions. Map-based visualisation allows users to understand the geographical position of a destination and its relationship to surrounding locations. Interactive maps can provide additional contextual information, including roads, landmarks, administrative areas, and other geographical features that may assist users in interpreting a location.

In customs operations, destination visualisation can help transform destination information contained in a customs declaration into a form that can be understood more rapidly by officers. Instead of requiring an officer to interpret a destination only as written text, a digital map can provide an immediate geographical representation of the declared location. This can be particularly useful when preparing field inspections, where understanding the location of a consignment may contribute to determining where operational activities should take place.

Within DouaneNav, destination visualisation constitutes one of the principal functions of the mobile application. The declared destination retrieved from the available declaration data is processed and represented on an interactive OpenStreetMap interface. The officer can therefore move from a textual destination contained in the declaration to a visual geographical representation of that destination. This functionality is intended to improve destination visibility and support the preparation and execution of customs field inspections without providing continuous GPS tracking of the vehicle.

1.1.2.6 Geocoding

Geocoding is the process of converting a textual description of a location, such as a place name, street address, or geographical description, into geographic coordinates, usually expressed as latitude and longitude. These coordinates allow a location to be positioned and represented on a digital map. Geocoding is therefore an important component of many mapping and location-based applications because it connects human-readable location information with geographic information systems.

In a customs context, a declared destination may initially be stored and presented as text within a declaration. For example, a destination may be identified by the name of a town, neighbourhood, or other locality. Before such a destination can be displayed accurately on an interactive map, the textual location must be associated with geographic coordinates. Geocoding provides the mechanism through which this conversion can take place.

Within DouaneNav, geocoding is used to transform the declared destination obtained from the declaration data into coordinates that can be displayed on the OpenStreetMap interface. The resulting coordinates allow the application to place a marker at the corresponding destination and provide the Customs Officer with a visual representation of the declared location. The accuracy of this representation depends on the quality and specificity of the destination information available in the declaration and on the geocoding service used by the application.

1.1.2.7 Customs Field Inspection

Customs field inspection refers to the physical examination and verification of goods, documents, vehicles, or other relevant elements by customs officers outside the primary administrative processing environment. It is carried out to verify the conformity of a consignment with the information declared to customs and to identify possible irregularities, such as discrepancies in the description, quantity, destination, or supporting documentation.

Field inspections require officers to have access to relevant information before and during an operation. Having clear information about the consignment, including its declared destination, transport details, and previous inspection information, can assist officers in preparing and documenting their activities more effectively.

Within DouaneNav, the field inspection function allows authorised Customs Officers to access relevant declaration information, view the declared destination, and record inspection outcomes directly through the mobile application. Officers can enter comments and attach photographs as supporting evidence. The resulting inspection report can be stored locally and subsequently synchronised with the appropriate customs system when an authorised connection is available.

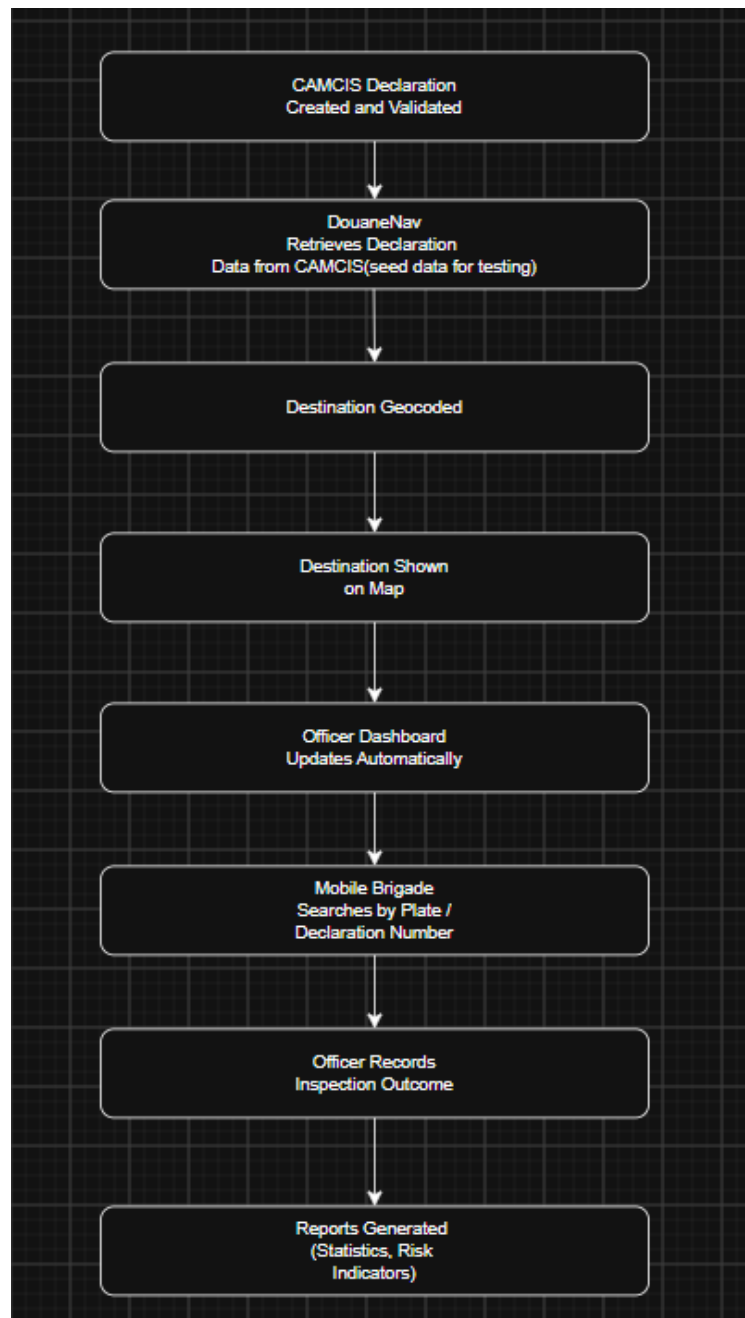


Figure 1: DouaneNav end-to-end operational workflow, from CAMCIS declaration to synchronised inspection report.

1.1.3 Theoretical Background

This study is grounded in **Decision Support System (DSS) theory**, first articulated by Anthony Gorry and Michael Scott Morton in 1971 (Power & D. J., 2007). The theory explains how computer-based systems can support organisational decision-making, particularly in situations involving structured and semi-structured decisions. In semi-structured situations, some aspects of the decision process can be automated, while the final judgement remains with the human decision-maker. This is relevant to DouaneNav because the retrieval and visualisation of shipment information can be automated, while the decision to conduct an inspection and the interpretation of inspection findings remain the responsibility of the Customs Officer.

The study also draws on **the theory of bounded rationality**, developed by Herbert Simon. Bounded rationality recognises that human decision-makers operate under limitations of time, information, and cognitive capacity. In the customs field environment, officers may need to process information from several documents while making operational decisions under time and connectivity constraints. DouaneNav applies this theory by reducing the amount of manual information processing required from officers. By retrieving relevant declaration information, presenting the declared destination visually, and providing structured inspection-recording tools, the system allows officers to focus their attention on inspection and decision-making rather than repetitive information retrieval and cross-referencing.

1.1.4 Contextual Background

Cameroon is served principally by the seaport of Douala, the country's main commercial gateway, and the deep-water port of Kribi, alongside land border posts and inland customs offices. Cameroon Customs (Direction Générale des Douanes, DGD) processes a substantial volume of import, export, and transit declarations daily, many of which relate to onward transit toward neighbouring landlocked countries such as Chad and the Central African Republic, using CAMCIS as its core declaration-processing platform. In this environment, officers are frequently deployed away from the point of declaration and rely on secondary information to decide where and when to inspect goods in transit.

Cameroon's growing smartphone and internet penetration, together with the increasing digitisation of trade documentation under CAMCIS, creates a practical opportunity to deploy a complementary, mobile-first decision-support tool of this kind. This study assumes that

customs declaration information will, in production, be available from CAMCIS, and is simulated locally for the purposes of development and evaluation. A system that retrieves declaration data, presents the result as an interactive map, and supports structured field inspection is both technically feasible within the current infrastructure and directly relevant to the operational needs of customs officers.

1.2 Statement of the Problem

Ideally, customs officers should be able to access the relevant information about a shipment from a centralised and easily accessible system, clearly identify its declared destination, and use this information to prepare and conduct inspections efficiently. The destination should be readily understandable without requiring officers to manually cross-reference multiple documents, while inspection findings should be recorded in a structured digital format and linked to the corresponding customs declaration for reliable follow-up and reporting.

For a single shipment, a customs officer may need to consult a considerable volume of information spread across several documents: the CAMCIS customs declaration, a bill of lading or transport document, a commercial invoice, a packing list, and transport company particulars. The destination of the goods exists somewhere within this set of documents, but it is not presented to the officer in an intuitive, visual way. As a result, officers spend time opening each document in turn, reading the information it contains, identifying where the goods are expected to go, and organising inspection activities accordingly. This process is manual, repeated for every consignment, and can slow down operational work, particularly at high-volume entry points.

A further consequence is that custom officers, who conduct inspections away from the point of declaration, often have limited advance visibility of the destination that goods are officially expected to reach. This reduces the efficiency of inspection planning. Additionally, after an inspection is completed, findings are often recorded through separate reporting methods, creating difficulties in maintaining a structured and synchronised link with the original declaration. Although CAMCIS provides automation for customs declaration processing, there is currently a need for a complementary decision-support solution that can retrieve declaration information, present declared destinations through interactive map visualisation, support structured field inspection with photographic evidence, and facilitate synchronisation of authorised inspection reports.

Therefore, this study focuses on the design and development of **DouaneNav**, a mobile-first decision support subsystem that complements CAMCIS by improving destination visibility, supporting customs field inspections, and enhancing the management of inspection information through digital reporting and synchronisation mechanisms.

1.3 Objectives of the Study

1.3.1 Main Objective

The main objective of this study is to design and develop DouaneNav, a mobile-first decision support subsystem connected to CAMCIS, whose mission is to retrieve declaration information, visualise declared destinations on an interactive map, assist Customs Officers during field inspections, and synchronise completed inspection reports back to CAMCIS, supported by a lightweight web dashboard for administration, without replacing CAMCIS itself.

1.3.2 Specific Objectives

- To design a mechanism for retrieving relevant customs declaration information, simulated locally during development, for use within the DouaneNav mobile application.
- To implement search functionality allowing officers to locate a declaration by declaration number, truck plate, importer, destination, or date.
- To visualise the retrieved destination on an interactive OpenStreetMap based map as a central, standalone feature of the mobile application.
- To implement an inspection module allowing Customs Officers to record inspection outcomes, including results, comments, and photographs, against a specific declaration.
- To design a synchronisation mechanism that stores completed inspection reports locally and transmits them back to CAMCIS once an authorised integration exists.
- To design a mobile dashboard providing alerts, recent inspection records, and synchronisation status to support monitoring of operational activities.
- To implement authentication and role-based access control (RBAC) distinguishing Administrators, Customs Officers, and Supervisors, and to test and evaluate the resulting system.

1.4 Research Questions

1.4.1 General Research Question

How can a mobile-first decision support subsystem, connected to CAMCIS, combine destination visualisation, field inspection support, and report synchronisation to improve customs officers' field operations in Cameroon?

1.4.2 Specific Research Questions

- How can customs declaration information be retrieved and presented within a mobile application to support destination verification by Customs Officers?
- How can an OpenStreetMap-based interactive map be used to visualise a declared destination in a way that supports Customs Officers during field operations?
- How can a mobile inspection module support Customs Officers in recording inspection results, comments, and photographs against a specific declaration?
- How can a mobile dashboard provide relevant operational information, including alerts, recent inspections, and synchronisation status, to authorised users?
- How can a synchronisation mechanism be designed to store completed inspection reports locally and support their transmission to CAMCIS when an authorised integration becomes available?
- How can authentication and role-based access control be implemented to provide appropriate access for Administrators, Supervisors, and Customs Officers?

1.5 Significance of the Study

1.5.1 To Customs Administration

The system offers customs administrators a centralised, auditable record of inspection activity and synchronisation outcomes, supporting more consistent field operations and reducing reliance on unstructured, paper-based inspection reporting.

1.5.2 To Customs Officers

By presenting destination information geographically and supporting structured, photograph-backed inspection recording, the system gives officers in the field improved visibility of where goods are declared to be going and a faster, more reliable way to document what they find.

1.5.3 To Importers and Clearing Agents

More efficient, better-targeted inspection planning can reduce unnecessary delays experienced by importers and clearing agents whose consignments would otherwise be subject to less-informed field checks.

1.5.4 To Researchers and the Academic Community

This study contributes to the growing body of research on mobile-first decision support tools for customs and trade-facilitation contexts in developing countries, and provides a model, including its bidirectional synchronisation approach, that future researchers in similar contexts can build upon.

1.6 Justification of the Study

CAMCIS already performs the processing of customs declarations in Cameroon and is expected to remain the country's principal customs platform. DouaneNav does not seek to replace CAMCIS; rather, it acts as a mobile-first companion subsystem connected to CAMCIS, whose mission is to retrieve declaration information, visualise destinations, support field inspections, and synchronise completed reports back to CAMCIS, functions that are not understood to fall within CAMCIS's current scope. Replacing or re-architecting a national platform such as CAMCIS would be a large-scale governmental undertaking beyond the scope of any single project; building a complementary subsystem alongside it is both more realistic and more immediately useful.

The increasing availability of mature, affordable mobile development frameworks and mapping platforms, together with the digitisation of declaration data already achieved through CAMCIS, makes this an appropriate moment to pursue such a subsystem. No comparable subsystem specifically targeting mobile destination visualisation, field inspection support, and report synchronisation, built to complement CAMCIS, currently exists within the Cameroonian customs context, which further supports the originality and relevance of this study.

1.7 Delimitation of the Study

1.7.1 Thematic Scope

This study is limited to the design and development of a mobile-first decision support subsystem that retrieves declaration information, visualises destinations on a map, supports structured field inspection, and synchronises completed inspection reports back to CAMCIS,

together with a lightweight companion web dashboard. It does not attempt to replace or reimplement CAMCIS's declaration-processing functions, nor does it provide continuous GPS-based vehicle tracking, payment processing, or automated risk-scoring. AI-powered document analysis is architected but not implemented within the present study; it is identified as a future-phase enhancement rather than a feature of the present system.

1.7.2 Geographical Scope

The application is designed with particular reference to customs operations in Cameroon, including the ports of Douala and Kribi and associated inland and transit routes. While the underlying approach could in principle be applied in other customs jurisdictions facing similar challenges, the design, development, and evaluation described in this study are limited to the Cameroonian context.

1.7.3 Time Scope

This study was conducted between June and August 2026. Background research and the Citymapper concept adaptation study were completed in June 2026, alongside the drafting of the Abstract, Chapter One, and Chapter Two. An internship at Cameroon Customs (Direction Générale des Douanes), Technical Service (Service Technique), Yaounde, took place between 22 July and 30 August 2026, during which further requirements validation and technical exposure informed the system's design. System design, development, and testing were carried out in parallel with, and following, the internship, with the final report submitted by 15 August 2026 and an oral defence scheduled for September 2026. Subsequent enhancements or large-scale deployment fall outside the scope of the study.

1.8 Organization of the Study

This report is organised into five chapters. Chapter One presents the introduction and background of the study, including the problem statement, objectives, research questions, significance, justification, and delimitation. Chapter Two presents a review of related literature covering theoretical, conceptual, and empirical perspectives, together with the project management plan and a presentation of the host institution.

Chapter Three describes the research methodology, including the research design, feasibility studies, population and sampling framework, and data collection and analysis methods. Chapter Four presents the implementation of the DouaneNav system and the results of its

testing and evaluation. Chapter Five presents the discussion of findings, conclusions, policy implications, and recommendations for future work.

CHAPTER TWO

LITERATURE REVIEW

This chapter reviews literature relevant to the design of DouaneNav as a mobile-first decision support subsystem that complements CAMCIS. It is organised into a theoretical review, a conceptual review, an empirical review structured around the specific objectives of the study, a presentation of the host institution, and the project management plan governing the execution of the study.

2.1 Theoretical Review

2.1.1 Decision Support System Theory

Decision Support System (DSS) theory provides a major theoretical foundation for DouaneNav. DSS theory explains how computer-based systems can support decision-makers by providing relevant information and analytical assistance, particularly when decisions are semi-structured and still require human judgement. In such situations, the system does not replace the decision-maker but assists in organising and presenting information required to reach an informed decision.

The relevance of this theory to DouaneNav lies in the nature of customs field operations. A Customs Officer may need to identify a consignment, retrieve its declaration information, determine its declared destination, and review previous inspection information before deciding how to proceed. DouaneNav supports these activities by bringing relevant information together, providing search functions, visualising the destination on an interactive map, and allowing inspection results to be recorded. The system therefore reduces the time required to obtain and organise information while leaving the final inspection decision to the Customs Officer.

The theory also supports the decision to develop DouaneNav as a subsystem rather than an independent replacement for CAMCIS. CAMCIS remains responsible for customs declaration processing, while DouaneNav provides additional information and field-support functions. This separation maintains the role of the human officer and positions the system as a tool for improving operational decision-making.

2.1.2 Bounded Rationality Theory

Bounded Rationality Theory, associated with Herbert Simon, proposes that decision-makers operate under limitations such as incomplete information, limited time, and restricted cognitive capacity. Instead of considering every possible piece of information before making a decision, individuals generally work with the information that is available and practical to process within the circumstances.

This theory is relevant to customs field operations because officers may work under time constraints while dealing with information distributed across several documents and systems. Manually locating a declared destination and comparing information from different sources can increase the effort required before an inspection decision can be made. These limitations become more significant when officers are operating away from fixed workstations or under variable connectivity conditions.

DouaneNav applies the principle of bounded rationality by reducing the amount of mechanical information-processing required from the officer. The system retrieves relevant declaration information, provides search mechanisms, presents the declared destination visually, and organises inspection records within a single mobile interface. By reducing unnecessary information-search and cross-referencing activities, the system allows officers to concentrate more on inspection, interpretation, and professional judgement.

Therefore, Bounded Rationality Theory provides a theoretical justification for the design of DouaneNav as an information-consolidation and visualisation tool. The system does not attempt to make decisions on behalf of Customs Officers; instead, it reduces some of the information and cognitive limitations under which those decisions are made.

2.1.3 Relevance of the Theories to DouaneNav

The two theories complement each other in explaining the design of DouaneNav. DSS theory establishes the role of the system as a tool that supports human decision-making, while Bounded Rationality Theory explains why such support is useful when officers face limitations in time, information, and cognitive capacity. Together, they justify the system's focus on

retrieving relevant declaration information, visualising destinations, organising inspection information, and supporting field officers without replacing their professional judgement.

2.2 Conceptual Review

2.2.1 Customs Digitisation and CAMCIS

Customs digitisation refers to the progressive replacement of paper-based procedures with electronic declaration, processing, and record-keeping. In Cameroon, this progression moved from PAGODE, to SYDONIA/ASYCUDA, to the current CAMCIS platform, each stage automating a greater share of the declaration lifecycle. CAMCIS represents the most advanced stage of this progression to date, but, as with most national single-window declaration platforms, its scope centres on the declaration itself rather than on field-level destination visualisation or structured, synchronisable inspection support.

2.2.2 Decision Support Systems

A Decision Support System (DSS) is a computer-based system designed to assist decision-makers by providing relevant information, analytical capabilities, and organised data to support decision-making. DSSs are particularly useful in situations where decisions involve a combination of available information, human judgement, and changing operational conditions. Rather than replacing the decision-maker, a DSS supports the user by reducing the time and effort required to obtain and interpret relevant information.

In the context of customs operations, a decision support system can assist officers by bringing together information required to assess and plan inspections. DouaneNav applies this concept by retrieving relevant declaration information, allowing officers to search for consignments, presenting declared destinations through interactive map visualisation, and providing access to previous inspection information. These functions reduce the effort required to manually locate and interpret information from several sources.

DouaneNav does not make the final decision concerning whether a shipment should be inspected or what action should be taken following an inspection. Instead, the system provides the information and tools required by the Customs Officer, while professional judgement remains with the officer. Therefore, the DSS concept provides the conceptual basis for DouaneNav's role as a decision-support tool rather than an autonomous decision-making system.

2.2.3 Mobile-First Field Support Systems

Mobile-first design refers to an approach in which the primary user experience is designed for mobile devices, taking into account their smaller screens, touch-based interaction, and variable network connectivity. Mobile-first systems are particularly useful in field operations because users may need to access information and perform tasks away from fixed workstations.

DouaneNav applies this concept by making the React Native mobile application the primary operational component of the system. Customs Officers can search declarations, view declared destinations on an interactive map, consult shipment information, and record inspection results directly from a mobile device. The web dashboard serves as a complementary interface for Administrators and Supervisors. This approach is appropriate because the main users of DouaneNav operate in field environments where rapid access to relevant information is required.

2.2.4 Geographic Information Visualisation and Geocoding

Geographic information visualisation refers to the representation of location-based information using maps and other spatial interfaces. It allows users to understand the geographical relationship between places more easily than when information is presented only as text. Geocoding is the process of converting a textual location, such as a city or neighbourhood name, into geographical coordinates that can be displayed on a digital map.

In DouaneNav, geocoding is used to convert a declared destination into geographical coordinates, which are then presented through an interactive OpenStreetMap interface. This allows Customs Officers to identify the location of a consignment at a glance and supports better preparation for field inspections. The system therefore applies geographic visualisation specifically to customs destination information rather than providing general navigation or continuous vehicle tracking.

2.2.5 Digital Customs Inspection and Reporting

Digital inspection refers to the use of electronic tools to record, store, and manage information collected during an inspection. Digital reporting allows inspection findings to be recorded in a structured format instead of relying on separate paper-based or informal reporting methods.

Such records can include inspection results, comments, photographs, dates, and other information associated with a specific declaration.

DouaneNav applies this concept through its mobile inspection module, which allows Customs Officers to record inspection outcomes, comments, and photographs against a specific declaration. The completed report is stored locally and prepared for synchronisation with CAMCIS when an authorised connection is available. This approach provides a structured digital record of field inspections and maintains a clear relationship between an inspection report and the corresponding customs declaration.

2.2.6 Complementary Systems and Bidirectional Synchronisation

A complementary system is designed to extend the value of an existing platform by consuming its data and providing functions that the original platform does not directly provide, without replacing or interfering with its core operation. This concept is central to DouaneNav, which retrieves declaration information from CAMCIS and separately synchronises completed inspection reports back to it. This bidirectional but asymmetric relationship, reading declarations and writing only inspection outcomes, allows DouaneNav to remain a complementary subsystem rather than a replacement for CAMCIS.

2.3 Empirical Review (Review by Objectives)

2.3.1 Retrieval of Customs Declaration Information

Existing customs digitisation initiatives demonstrate the value of making declaration information available electronically. In Cameroon, the CMR Customs Declaration App allows users to prepare customs declarations electronically and supports offline creation before submission through electronic declaration terminals. The application therefore demonstrates the use of mobile technology to collect and transmit customs-related declaration information rather than relying entirely on paper-based procedures (Cameroon Customs, n.d.).

Similarly, CAMCIS provides the national platform through which customs declarations are entered, stored, and processed. The Cameroon Economic Policy Institute (CEPI, 2024) documents that customs declarations are entered and validated within CAMCIS, confirming that structured declaration information is already available within the national customs information environment.

However, these existing solutions primarily address the creation, submission, and processing of declarations. They do not provide a dedicated mobile field-support subsystem that retrieves relevant declaration information for Customs Officers, presents the declared destination visually, and links the retrieved information directly to subsequent inspection activities.

DouaneNav builds on this existing digital foundation by proposing a complementary retrieval mechanism through which relevant declaration information can be accessed within the mobile application. During development, this mechanism is demonstrated using a simulated dataset representing CAMCIS records, while direct retrieval from CAMCIS is reserved for an authorised future integration. The contribution of DouaneNav is therefore not to create another declaration-processing platform, but to reuse existing declaration information for mobile field operations and inspection support.

2.3.2 Declaration Search Functionality

Search and retrieval functions are commonly used in digital information systems to allow users to locate specific records without manually examining an entire dataset. In customs and logistics environments, electronic systems use identifiers such as declaration numbers, shipment references, and other consignment information to retrieve relevant records more efficiently than manual document searches.

Existing customs platforms such as CAMCIS provide structured declaration records that can be identified and processed electronically. However, the availability of structured declaration records does not by itself provide a dedicated field-oriented search interface designed around the needs of Customs Officers. In particular, officers conducting field inspections may need to identify a consignment using information immediately available to them, such as a declaration number or truck registration plate.

DouaneNav extends this principle by providing a mobile search interface through which officers can locate declarations using multiple identifiers, including declaration number, truck plate, importer, destination, and date. This multi-criteria approach is intended to reduce the time required to locate a consignment before an inspection. Unlike a general declaration-processing search, the search function in DouaneNav is directly connected to destination visualisation and inspection history, allowing the retrieved declaration to proceed immediately into the field-inspection workflow.

The contribution of DouaneNav therefore lies in combining multi-criteria declaration retrieval with the subsequent map visualisation and inspection functions within a single mobile field-support application.

2.3.3 Destination Visualisation

Map-based visualisation is widely used in navigation, logistics, and field-operation systems to represent geographical information in a form that can be interpreted more quickly than textual location descriptions. Applications such as Citymapper demonstrate how location information can be presented through interactive maps to help users understand destinations and routes. In logistics and transport operations, similar geographic visualisation approaches can support the identification and management of locations associated with goods and vehicles.

In the customs context, however, the declared destination is primarily recorded as part of declaration and transport documentation. The Cameroon Economic Policy Institute (CEPI, 2024) describes the documentation involved in Cameroonian trade procedures, showing that information concerning goods and their destination is distributed across several documents. This creates an opportunity to improve how destination information is presented to officers during field operations.

DouaneNav applies map-based visualisation specifically to customs declarations by converting the declared destination into geographical coordinates through geocoding and displaying it on an interactive OpenStreetMap interface. The map is connected directly to the retrieved declaration, allowing an officer to move from identifying a consignment to understanding its declared destination within the same application.

The main contribution of DouaneNav is therefore the adaptation of map-based visualisation to customs field inspection support. Unlike general navigation applications, DouaneNav does not aim to provide continuous vehicle navigation or GPS tracking. Its purpose is to provide Customs Officers with a clear geographical representation of the destination declared for a specific consignment, thereby supporting inspection preparation and operational awareness.

2.3.4 Digital Inspection Recording

Digital inspection systems are increasingly used to replace unstructured paper-based recording with electronic records that can be stored, retrieved, and associated with specific operational

activities. Such systems may allow officers or inspectors to record findings, observations, photographs, and other evidence directly during field operations. Digital recording can improve the consistency and traceability of inspection information by maintaining the collected information in a structured format.

Within customs operations, inspection activities are linked to the verification of goods and their accompanying documentation. However, the information reviewed in the existing customs documentation does not itself provide a dedicated mobile interface for recording field inspection outcomes together with photographic evidence and comments against a specific declaration.

DouaneNav addresses this gap through a mobile inspection module linked directly to the retrieved declaration. Customs Officers can record the inspection result, add comments, and capture photographs as evidence. The information is stored as a structured inspection report associated with the relevant consignment.

The contribution of DouaneNav is therefore the integration of inspection recording with declaration retrieval and destination visualisation in one mobile workflow. Rather than treating inspection reporting as a separate activity, the system links the inspection outcome to the declaration that the officer retrieved and reviewed before carrying out the field inspection.

2.3.5 Inspection Report Synchronisation

Data synchronisation mechanisms are used in distributed and mobile systems to transfer information between local devices and central servers. They are particularly important in mobile environments where network connectivity may be unavailable or unreliable. A common approach is to temporarily store information locally and transmit it when a connection becomes available, while retaining failed records for later retry.

In customs and other field operations, this approach is relevant because officers may complete inspections in locations where reliable connectivity cannot always be assumed. A field application that depends entirely on continuous connectivity may prevent officers from completing reports when communication with the central system is temporarily unavailable.

DouaneNav applies this principle by storing completed inspection reports locally before attempting synchronisation. Once an authorised connection to CAMCIS is available, the

reports can be transmitted to the central system. If synchronisation fails, the report remains stored locally and is placed back in the synchronisation process for automatic retry rather than being deleted.

The contribution of DouaneNav is the combination of local inspection-report storage, controlled transmission, and automatic retry within a customs-specific workflow. The mechanism is designed as a future authorised integration with CAMCIS rather than claiming that direct live CAMCIS synchronisation was implemented during this academic project. This distinction ensures that the system's demonstrated functionality is not confused with a production integration that requires official authorisation and access.

2.4 Presentation of the Enterprise (Internship)

2.4.1 Presentation of the Internship

The internship was carried out at the Cameroon Customs (Direction Générale des Douanes), Yaounde, Cameroon, within the Technical Service (Service Technique), from 22 July to 30 August 2026. The service is involved in the technical support, monitoring, and modernisation of customs operations, including support related to digital systems such as CAMCIS.

2.4.2 Activities Carried Out

During the internship, the main activities included technical support, installation and configuration of software, monitoring of surveillance systems, research into Cameroon Customs and its information systems, and development work related to the DouaneNav project.

2.4.3 Internship Experience

The internship provided practical exposure to the technical environment of Cameroon Customs and enabled interaction with personnel from different services. It also provided an opportunity to relate software engineering concepts to a real institutional environment and to better understand the role of digital systems in customs operations.

2.4.4 Strengths, Weaknesses and Problems Encountered

The internship provided a structured working environment, technical guidance, and practical exposure to customs information systems. Challenges included adapting to customs

terminology and procedures, understanding the institutional environment, and translating theoretical knowledge into practical development work.

2.4.5 Recommendations

The internship experience suggests the need for continued digitalisation of internal customs processes and the progressive replacement of outdated computer equipment to support secure and efficient use of modern digital systems.

2.5 Presentation of the Project Management Plan

2.5.1 Purpose of the Plan

This project management plan defines how the design and development of DouaneNav will be carried out, monitored, and completed within the time available for the study. It establishes the scope of work, the assumptions and constraints under which the project operates, and the schedule against which progress will be measured.

2.5.2 Executive Summary

DouaneNav is a mobile-first decision support subsystem intended to complement CAMCIS by retrieving declaration information, visualising the result on an interactive map, supporting structured field inspection, and synchronising completed inspection reports back to CAMCIS, supported by a lightweight web dashboard. The project is carried out as an academic requirement over approximately three months, from June to September 2026, progressing from background research and conceptual design in June, through an internship-informed design and development phase between July and mid-August, to final report submission on 15 August 2026 and an oral defence in September 2026.

2.5.3 Assumptions and Constraints

2.5.3.1 Assumptions

- Declaration information equivalent to what is held in CAMCIS will be accessible for development and testing purposes through a representative mock dataset designed to simulate CAMCIS records, since real CAMCIS data will not be used.
- The OpenStreetMap will remain accessible and within free or low-cost usage tiers for the duration of development and testing.

- The field supervisor has provided, and will continue to validate, empirical literature sources and internship-specific content in time for inclusion in the final report.
- Customs Officer users will have access to a smartphone capable of running the mobile application in the field.

2.5.3.2 Constraints

- The project must be completed by the institution's submission deadline of 15 August 2026, with an oral defence scheduled for September 2026.
- Direct API or database-level integration with the live CAMCIS platform is not available for this student project; the system will therefore be developed and demonstrated using a simulated dataset designed by the student to stand in for CAMCIS records, a limitation to be stated explicitly in the methodology.
- Development is carried out by a single student within the available academic schedule, alongside the internship placement running concurrently from 22 July to 30 August 2026.
- AI-powered document analysis is architected but not implemented within the present study, and is deferred to a future phase.

2.5.4 Scope Management

2.5.4.1 Work Breakdown Structure

The project is broken down into the following major phases of work:

- Requirements analysis and background research, including the Citymapper concept adaptation study.
- Conceptual, organisational, and functional design of the DouaneNav subsystem.
- Backend development, including a simulated dataset designed to stand in for CAMCIS declaration data.
- Mobile application development, including search, destination map, and inspection modules.
- Development of the synchronisation mechanism, including queueing and automatic retry.

- Web dashboard development for Administrator and Supervisor use.
- System testing, evaluation, and documentation of results.
- Final report writing and preparation for oral defence.

2.5.4.2 Deployment Plan

The mobile application will initially be tested on physical or emulated Android/iOS devices in a local development environment, using a simulated dataset designed by the student to stand in for CAMCIS declaration records. Where feasible, a demonstration deployment of the backend and web dashboard may be hosted on a cloud platform to allow the field supervisor and academic supervisor to access and evaluate the system remotely. A full production deployment integrated with the live CAMCIS platform falls outside the scope of this study and is identified as a direction for future work.

2.5.4.3 Change Control Management

Any change to the scope of the system, such as the addition or removal of a functional objective, will be documented and submitted to the field supervisor and academic supervisor for validation before implementation, in line with the phased, supervisor-validated approach already followed for the Citymapper analysis and concept adaptation stages of this project.

2.5.4.4 Report (Project Summary, Milestones, Working Days)

The overall project timeline, expressed as a set of milestones with target periods and expected deliverables, is presented in the table below, reflecting the project's actual compressed schedule between June and September 2026.

Table 1 : project timeline

Milestone	Target Period	Deliverable
M1 — Project initiation and topic definition	Early June 2026	Approved project topic
M2 — Citymapper analysis and concept adaptation report completed	June 2026	Citymapper analysis report; concept adaptation report
M3 — Abstract, Chapter One and Chapter Two drafted	June 2026	Draft Abstract, Chapter One, Chapter Two

M4 — Internship placement, Technical Service, DGD Yaounde	22 July – 30 August 2026	Field exposure, requirements validation, technical skills
M5 — System design and backend development	Late July 2026	Data model, workflow design, backend implementation
M6 — Mobile app, map integration and CAMCIS-data retrieval mechanism integrated	Late July – early August 2026	Working mobile app; mock CAMCIS-data retrieval; map visualisation
M7 — Inspection module, synchronisation mechanism and web dashboard completed	Early August 2026	Inspection recording, sync queue/retry, admin dashboard
M8 — System testing and evaluation	Early to mid-August 2026	Test results, evaluation
M9 — Final report compiled and submitted	15 August 2026	Complete project report
M10 — Oral defence	September 2026	Project defence

2.5.5 Schedule and Time Management

2.5.5.1 Milestones Management

Each milestone listed above marks the completion of a distinct phase of work and is reviewed with the relevant supervisor, academic or field, before the next phase begins, consistent with the phased validation approach already established for this project.

2.5.5.2 Schedule Management

2.5.5.2.1 Dependencies and Activity Sequencing

Backend development and the simulated CAMCIS-data mechanism must be completed before mobile application development can be meaningfully tested, since the mobile app depends on having declaration data available to retrieve and display. Map visualisation depends on the backend correctly geocoding a retrieved destination. The synchronisation mechanism depends on the inspection module being functional, since it operates on completed inspection reports. Web dashboard development can proceed in parallel with mobile development, provided

integration points are agreed in advance. Development work runs concurrently with, and is informed by, the internship placement between 22 July and 30 August 2026.

2.5.5.3 Schedule and Time Management Report

2.5.5.3.1 Network Diagram Report

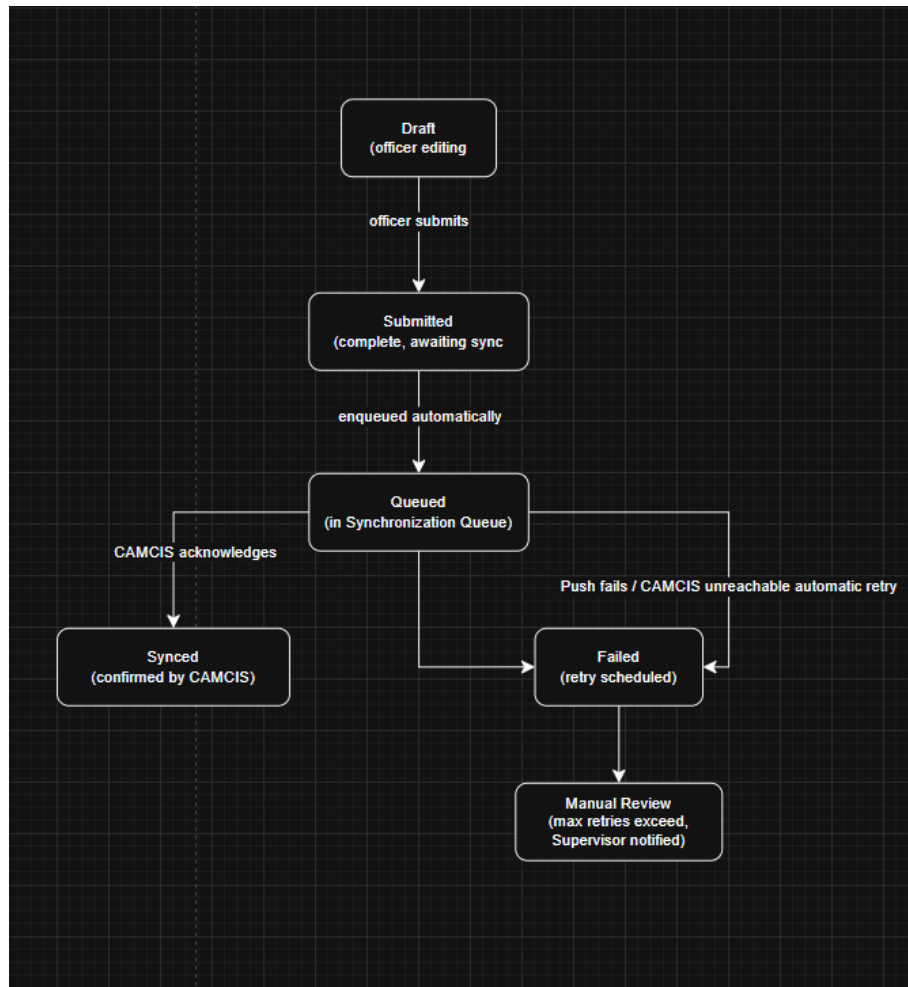


Figure 2: Report synchronisation lifecycle, illustrating the dependency between inspection completion and synchronisation.

Requirements analysis and design lead into backend development, which in turn enables mobile application development (search, map, inspection modules); the synchronisation mechanism depends on the inspection module; web dashboard development runs in parallel with mobile development and converges with it at the testing phase; testing and evaluation lead into final documentation and defence. The path from requirements and design through backend development, mobile development, synchronisation, testing, and final reporting constitutes the critical path, since any delay along this path directly delays project completion, whereas web dashboard development has schedule slack.

2.5.5.3.2 Top Level Task Report

At the top level, the project consists of four major streams of work: research and design, internship-informed development, testing and evaluation, and final documentation, corresponding respectively to milestones M1 to M3, M4 to M7, M8, and M9 to M10 in the table above.

2.5.5.3.3 Critical Task Report

The mobile application's core modules (search, map, inspection) and the synchronisation mechanism represent the critical tasks of the project, since dashboard testing and final evaluation depend on their successful completion. Given the compressed schedule between June and September 2026, delays in either task are expected to have the greatest impact on the overall project timeline and should therefore be prioritised and monitored most closely.

CHAPTER THREE

METHODOLOGY

This chapter describes the methodology followed in designing and developing DouaneNav. As the study focuses on designing a working software application the methodology is adapted to a systems-development context.

3.1 Research Design

This study adopts a **Research Design Methodology** appropriate for the systematic design and development of the DouaneNav. The methodology provides a structured approach for identifying the problem, collecting and analysing relevant information, defining system requirements, designing the proposed solution, implementing the system, and evaluating the resulting application.

The study combines documentary research with software engineering practices. Documentary research was used to understand Cameroon Customs procedures, CAMCIS, customs digitisation, and the operational requirements relevant to the project. The information obtained was then translated into functional and technical requirements for DouaneNav.

The development process consisted of requirements analysis, system design, implementation, testing, and evaluation. The Agile methodology described in Section 3.3 was used to guide the iterative development of the system.

This research design was selected because it provides a systematic link between the identified operational problem and the development of a practical technological solution. It also allows the resulting system to be evaluated according to the functional requirements and objectives established in Chapter One.

3.2 Feasibility Studies

3.2.1 Market Study

As established in Chapter One's Justification (Section 1.6), no comparable mobile-first subsystem targeting destination visualisation and field-inspection support, built to complement CAMCIS, currently exists within the Cameroonian customs context. This supports the market case for the project, though it is worth noting the study does not involve a formal commercial

market analysis, since DouaneNav is intended for internal operational use by Cameroon Customs rather than as a commercial product.

3.2.2 Environmental Feasibility

Cameroon's growing smartphone and internet penetration supports the feasibility of a mobile-first tool. At the host institution specifically, the working environment was found to be orderly and secure (Section 2.4.4), though internal digitalisation remains limited and some computers run outdated operating systems such as Windows 7, an environmental constraint relevant chiefly to any future desktop-based component of the system rather than to the mobile application itself.

3.2.3 Technical Feasibility

The technologies selected, React Native for the mobile application, React.js for the web dashboard, Node.js with Express for the backend, and MongoDB for the database, are mature, well documented, and widely used, reducing technical risk. The OpenStreetMap provides reliable geocoding and map rendering. Because direct CAMCIS integration is not available to the student, technical feasibility during development depends on the use of a simulated dataset, considered a reasonable and common approach for student projects working with secured government platforms.

3.2.4 Economic Feasibility

Development relies primarily on free or low-cost tiers of the technologies involved (open-source frameworks, and free usage tiers of the OpenStreetMap), with the main cost being the student's development time rather than licensing or infrastructure expenditure. A detailed cost estimate is presented in Chapter Five.

3.2.5 Socio-Political Feasibility

The project aligns with Cameroon Customs' own stated modernisation mandate under Decree No. 2013/066 (Section 2.3), and with the broader trajectory of customs digitisation already demonstrated by the PAGODE-to-SYDONIA-to-CAMCIS progression (Section 1.1.1), suggesting reasonable socio-political feasibility, though formal institutional adoption would require approval beyond the scope of this academic study.

3.2.6 Feasibility Report

Taken together, the feasibility studies above indicate that DouaneNav is feasible as an academic project within the stated constraints: technically achievable with mature, low-cost tools; economically modest in cost; environmentally supported by growing mobile penetration, notwithstanding internal digitalisation gaps at the host institution; and socio-politically aligned with the customs administration's own modernisation objectives.

3.3 System Development Methodology

Software development methodologies provide structured approaches for analysing, designing, implementing, testing, and maintaining software systems. For the development of DouaneNav, three commonly used methodologies were considered: Waterfall, V-Model, and Agile.

3.3.1 Waterfall Methodology

The Waterfall methodology follows a sequential development process in which each phase is completed before the next phase begins. The main phases include requirements analysis, system design, implementation, testing, deployment, and maintenance.

The main advantage of Waterfall is its structured and well-documented process, which makes project planning and progress tracking straightforward. However, it provides limited flexibility when requirements change during development. This limitation makes it less suitable for DouaneNav because some system requirements were refined progressively during the design and development stages.

3.3.2 V-Model Methodology

The V-Model is an extension of the Waterfall approach that places strong emphasis on verification and validation. Each development phase is associated with a corresponding testing phase. For example, system requirements are linked to acceptance testing, while system design is linked to system testing.

The V-Model provides strong quality control and is suitable for projects where requirements are clearly defined from the beginning. However, like Waterfall, it is relatively rigid and makes changes to requirements more difficult once development has progressed. This reduces its suitability for DouaneNav, where modules were developed and refined progressively.

3.3.3 Agile Methodology

Agile methodology focuses on iterative development, continuous improvement, regular testing, and adaptation to changing requirements. Instead of developing the entire system at once, the system is divided into smaller functional components that can be developed, tested, evaluated, and improved progressively.

Agile is suitable for DouaneNav because the project consists of several interconnected modules, including authentication, declaration search, destination visualisation, inspection management, synchronisation, and web dashboard management. Each module could be developed and tested progressively before being integrated with the rest of the system.

Another advantage of Agile is its flexibility. Feedback received during development could be incorporated without requiring the entire project to be redesigned. This was particularly useful because the project involved continuous refinement of system requirements and adaptation of the application to the customs operational context.

Although Agile is commonly associated with collaborative development teams and frequent stakeholder meetings, its iterative principles can also be applied to an individual academic software project through progressive development, testing, evaluation, and refinement.

3.3.4 Comparison of Development Methodologies

Table 2: Comparison of methodologies

Criteria	Waterfall	V-Model	Agile
Development approach	Sequential	Sequential with integrated testing	Iterative and incremental
Flexibility	Low	Low	High
Requirement changes	Difficult	Difficult	Easier
Testing	Mainly after implementation	Integrated throughout development	Continuous
User feedback	Limited	Limited	Frequent
Suitability for DouaneNav	Moderate	Moderate	High

3.3.5 Choice of Methodology

The three methodologies were considered in relation to the requirements, development environment, and constraints of the DouaneNav project. Waterfall and V-Model provide strong structure and documentation but are less flexible when requirements evolve during development. Agile provides greater flexibility by allowing features to be developed, tested, reviewed, and refined progressively.

DouaneNav was developed as a modular system consisting of several interconnected components. Development therefore required continuous testing and refinement as individual modules were completed and integrated. The project also involved changing requirements and progressive validation during development.

Based on the characteristics described, Agile was chosen because it provides the flexibility, iterative development, continuous testing, and progressive refinement required for the development of DouaneNav within the available academic timeframe and project constraints.

3.4 Functional Requirements

Functional requirements describe the services that DouaneNav must provide to its users. The system was designed to satisfy the operational requirements of Cameroon Customs by allowing authorized personnel to securely access declaration information, verify destinations, perform inspections, and manage customs operations. They are

- User Authentication
- Dashboard Access
- Declaration Search
- Declaration Details
- Destination Visualization using OpenStreetMap
- Inspection Management
- Inspection Report Generation
- AI Document Analysis Prototype (Future Enhancement)
- Data Synchronization
- User Management (Administrator only)

3.4.1 Role Played by Each Actor

Administrator

- Manage users
- View all declarations
- Monitor synchronization
- Access reports
- Configure the system

Supervisor

- Review inspections
- View reports
- Monitor officers
- Access dashboard

Customs Officer

- Search declarations
- Verify destination
- Perform inspection
- Create inspection reports
- View AI analysis results from the prototype module

3.5 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics of the DouaneNav system. They describe how the system should perform its functions in terms of security, usability, reliability, performance, maintainability, scalability, and compatibility.

The following non-functional requirements were considered during the design and implementation of DouaneNav:

- **Security:** The system shall protect user accounts through authentication, password hashing, JWT-based sessions, and role-based access control.

- **Usability:** The mobile application shall provide a simple interface that allows Customs Officers to search declarations, view destinations, and record inspections with minimal interaction.
- **Performance:** The system shall return declaration search results and display relevant system information within a reasonable response time under normal operating conditions.
- **Reliability:** Inspection reports stored locally shall not be deleted when synchronisation fails and shall remain available for subsequent synchronisation attempts.
- **Availability:** The system shall allow inspection information to remain accessible locally when network connectivity is temporarily unavailable.
- **Maintainability:** The system shall use a modular architecture that allows individual components to be maintained or extended independently.
- **Scalability:** The system architecture shall allow additional users, declarations, inspection records, and future integrations to be accommodated without redesigning the entire system.
- **Compatibility:** The mobile application shall operate on supported Android devices, while the administrative dashboard shall be accessible through a modern web browser.

3.6 Population of the Study

The population relevant to this study consists of personnel within Cameroon Customs who may use or interact with a digital system for customs information retrieval, destination verification, field inspection, and inspection reporting. This includes Customs Officers, Supervisors, and Administrators.

3.6.1 Target Population

The target population consists of personnel of the Cameroon Customs (Direction Générale des Douanes), particularly staff whose activities are related to customs operations, technical support, supervision, and digitalisation.

3.6.2 Accessible Population

The accessible population was limited to personnel encountered during the internship at the Technical Service of the DGD headquarters in Yaounde. The internship provided an

opportunity to observe the working environment, technical activities, and general use of digital systems within the institution.

3.7 Sampling Technique

Purposive sampling was used to select relevant documentary sources and institutional information for the study. Sources were selected based on their relevance to Cameroon Customs procedures, customs digitisation, CAMCIS, and the requirements of the proposed DouaneNav system.

No statistical sample was required because the main purpose of the study was the design and development of a software solution rather than the measurement of a population's characteristics.

3.8 Data Collection Methods

Data was collected using observation and documentary review.

3.8.1 Observation

Observation was conducted during the internship at the Technical Service of the Cameroon Customs (Direction Générale des Douanes). The observation focused on the institutional working environment, technical activities, existing digital systems, and general operational practices relevant to the development of DouaneNav.

The observations helped provide practical context for the system design and supported the identification of technical and operational considerations. However, observations from the Technical Service were not used to make unsupported claims about front-line declaration processing. Specific customs procedures were therefore verified using documentary sources.

3.8.2 Documentary Review

Documentary review was used to obtain detailed information about customs procedures, customs digitisation, CAMCIS, and the institutional responsibilities of Cameroon Customs. The reviewed materials included official Cameroon Customs and CAMCIS documentation, government regulations, UNCTAD documentation on customs automation, the Cameroon

Economic Policy Institute (CEPI) report on import and export procedures in Cameroon, and relevant academic and theoretical literature.

The documentary findings were used to establish the problem context, identify system requirements, and guide the design and development of DouaneNav.

3.9 Validity and Reliability

3.9.1 Validity

Validity was supported through the use of multiple relevant sources and methods. Information obtained through observation during the internship was compared with information obtained from official documentation and independent documentary sources. This helped ensure that the requirements and problem context used in the study were consistent with the available evidence.

3.9.2 Reliability

Reliability was supported by using documented and verifiable sources, together with a consistent observation approach during the internship. The same documented sources can be consulted by other researchers, while the observations were recorded in relation to the technical and operational context relevant to the study.

3.10 Method of Data Analysis

Documentary analysis was applied to the secondary sources described in Section 3.6.2, identifying and synthesising the specific passages describing customs declaration verification procedures, in particular the practice of cross-referencing a declaration against its accompanying bill of lading, invoice, and packing list, which directly informed the problem statement and functional requirements presented in Chapter One. This analysis is fully traceable to named, citable sources (Section 2.3) rather than to the researcher's own field notes or recollection. For system evaluation, standard software testing methods, unit testing of individual functions, functional testing of each module, and informal user-acceptance review with the field supervisor, are applied and reported in Chapter Four.

3.10.1 Results of Documentary Data Analysis

The documentary analysis conducted from the selected secondary sources provided the evidence used to define the problem addressed by this study and derive the functional requirements of DouaneNav. The reviewed documents were analysed to identify information related to customs declaration processing, digitalisation of customs operations, and the requirements for improving access to shipment information during field activities.

The analysis produced the following findings:

Table 3: Results of documentary analysis

Source	Evidence Identified	Contribution to DouaneNav Design
Cameroon Economic Policy Institute (CEPI, 2024) report on import and export procedures in Cameroon	Customs clearance procedures require verification of information contained in multiple supporting documents, including transport documents, commercial invoices, and packing lists accompanying the customs declaration.	Supports the need for a system that retrieves declaration information and presents important shipment details in a more accessible manner.
Cameroon Customs and CAMCIS documentation	The CAMCIS platform provides electronic management of customs declarations and contributes to the digital transformation of customs operations.	Supports the development of DouaneNav as a complementary subsystem that works alongside CAMCIS rather than replacing it.
UNCTAD documentation on ASYCUDA customs systems	Modern customs environments require digital information management, data exchange, and improved accessibility of customs information.	Supports the implementation of synchronisation mechanisms and digital inspection reporting.
Decree No. 2013/066 of 28 February 2013 relating to the	Defines the responsibilities and organisational structure of Cameroon Customs administration.	Provides the institutional context for identifying the intended users of the

organisation of Cameroon Customs		system, including Customs Officers, Supervisors, and Administrators.
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The findings obtained from this documentary analysis directly informed the system requirements presented in Section 3.3. In particular, the identified need for improved access to declaration information, destination verification, and structured inspection reporting guided the development of the main modules of DouaneNav, including Declaration Management, Destination Visualisation, Inspection Management, and Synchronisation.

3.11 Method of Project Analysis

The analysis and design of DouaneNav were carried out using standard software engineering modeling techniques. Before implementation, the system was represented using Unified Modeling Language (UML) diagrams and an Entity Relationship Diagram (ERD). These models provided a visual representation of the system requirements, user interactions, workflows, object relationships, and database structure. They also served as a blueprint during implementation by ensuring that every functional requirement identified during the analysis phase was correctly translated into software components.

The following subsections present the design models developed for the project.

3.11.1 Use Case Diagram

The Use Case Diagram provides a high-level overview of the interactions between the users of DouaneNav and the system. It identifies the different actors involved and the services each actor can perform according to their responsibilities.

The system has three main actors:

- Administrator
- Supervisor
- Customs Officer

Each actor interacts with specific system functions such as authentication, declaration management, destination verification, inspection management, report generation, synchronization, and user administration.

The diagram helps define the functional boundaries of the system and ensures that every user role has access only to the operations assigned to it.



Figure 3: Use Case Diagram

3.11.2 Sequence Diagram

Sequence Diagrams describe the interaction between system components over time. They illustrate the chronological order in which requests are exchanged between the user interface, backend services, business logic, and database.

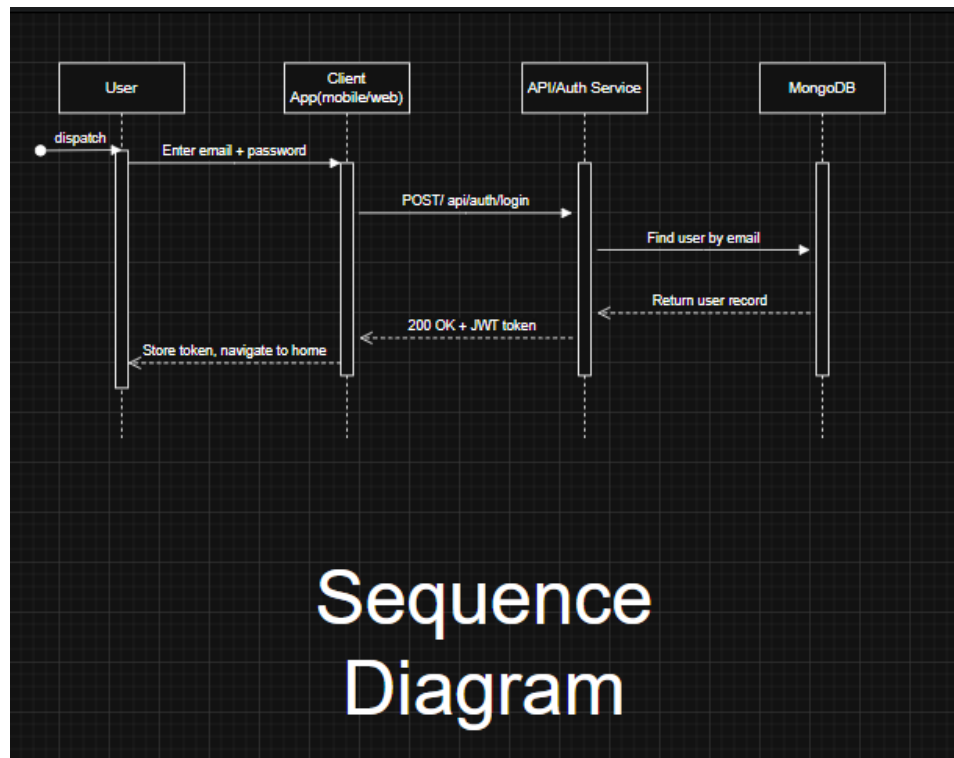


Figure 4: Login Sequence Diagram

3.11.3 Class Diagram

The Class Diagram illustrates the static structure of DouaneNav by showing the major classes that make up the application, together with their attributes, methods, and relationships.

The diagram includes the principal entities used during development, such as:

- User
- Declaration
- Inspection
- Report
- AIAnalysis
- SynchronizationLog

It also illustrates the relationships between the service layer, repositories, controllers, and domain models developed using the Clean Architecture adopted for the project.

The Class Diagram served as the blueprint for implementing the backend components and organizing the application into reusable and maintainable modules.

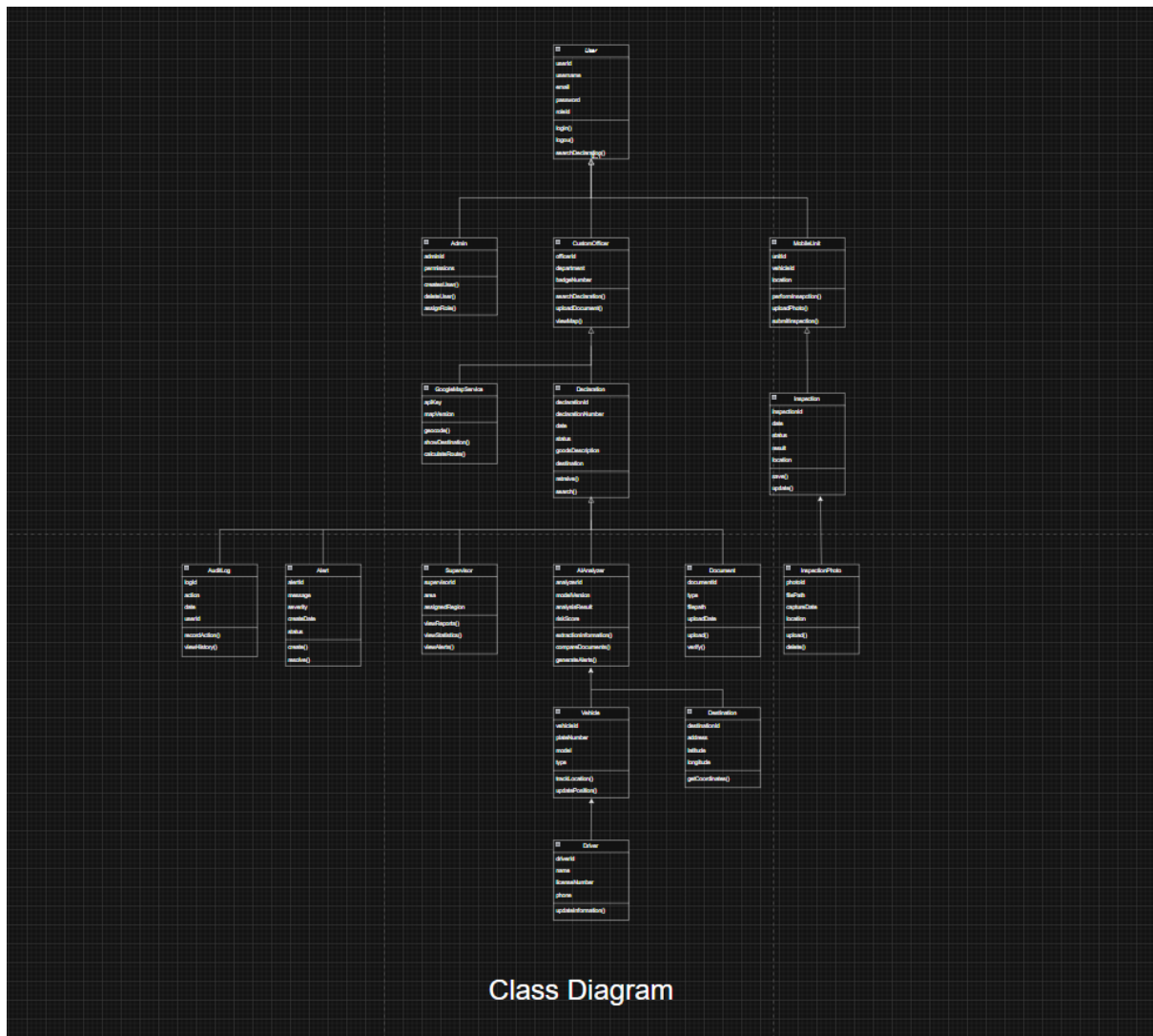


Figure 5: Class Diagram

3.11.4 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram illustrates the logical organisation of MongoDB collections and the relationships between the data entities used by the system.

The principal entities represented include:

- Users
- Declarations
- Inspections
- Reports
- AIAnalysis
- SynchronizationLogs

- AuditLogs

The ERD was developed to ensure data consistency, minimize redundancy, and facilitate efficient retrieval of customs declaration information during system operation.

Although MongoDB is a NoSQL database, the ERD provides a conceptual representation of how the collections are related and how information flows throughout the application.

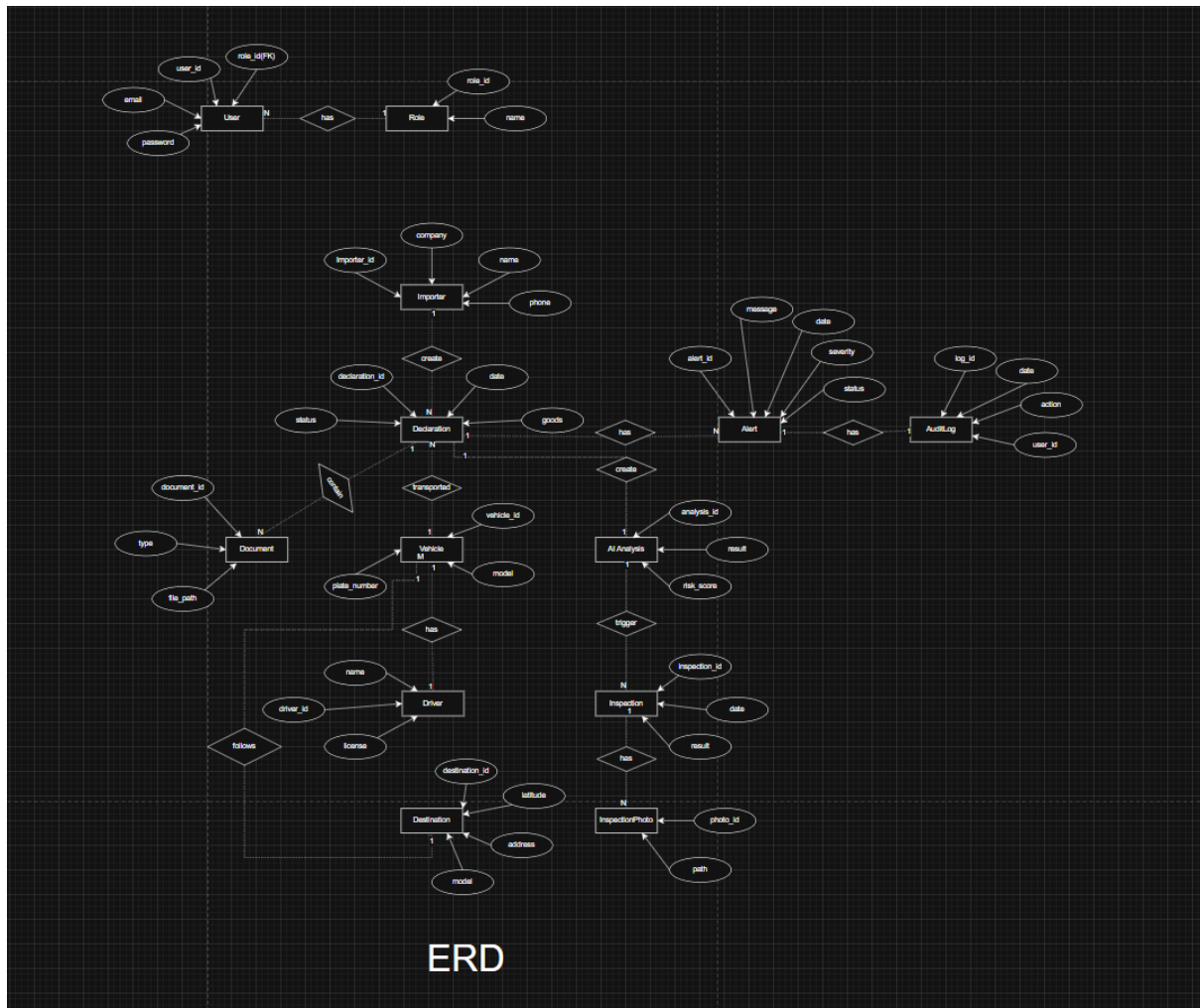


Figure 6: Entity Relationship Diagram (ERD)

3.12 Data Dictionary

The data dictionary provides a structured description of the main data elements used in the DouaneNav system. It defines the principal entities, their purpose, and the type of information they contain. The dictionary was used to guide the design of the database, backend services, mobile application, and web dashboard.

Table 4: Data Dictionary

Data Element	Description	Main Information Stored
User	Represents a person authorised to access DouaneNav.	User ID, name, email, password, role, account status
Role	Defines the permissions assigned to a system user.	Administrator, Supervisor, or Customs Officer
Declaration	Represents a customs declaration retrieved from the simulated CAMCIS dataset.	Declaration number, importer, truck plate, driver, transport company, destination, date, status
Destination	Represents the declared destination of a shipment.	Destination name, address, latitude, longitude
Inspection	Represents an inspection performed against a specific declaration.	Inspection ID, declaration ID, officer, result, comments, photographs, date, location
Photograph	Represents photographic evidence attached to an inspection.	Photograph ID, inspection ID, file path or URI, date
Inspection Report	Represents the structured result of an inspection.	Report ID, inspection ID, findings, comments, status, creation date
Synchronization Record	Records the transmission status of an inspection report to CAMCIS.	Synchronization ID, report ID, status, attempt count, error information, timestamp
Audit Log	Records important system activities for accountability and monitoring.	Log ID, user, action, timestamp, affected record
Search Criteria	Represents information entered by an officer to locate a declaration.	Declaration number, truck plate, importer, destination, or date
Map Coordinates	Represents the geographical position used to display a destination.	Latitude and longitude

The data dictionary reflects the information required to support the functional objectives of DouaneNav. Declaration data is designed to represent information that would be retrieved from

CAMCIS once an authorised integration becomes available, while inspection and synchronisation data are generated within DouaneNav during field operations.

3.13 Materials Used

The design and development of DouaneNav required a combination of hardware, internet resources, development frameworks, software tools, and supporting technologies. The main materials used during the project are summarised below.

3.13.1 Hardware Materials

The principal development computer used for the project was an HP ENVY Laptop 17-cr0xxx with the following specifications:

Table 5: Hardware Materials

Component	Specification
Manufacturer	HP
Model	HP ENVY Laptop 17-cr0xxx
Operating System	Windows 11 Home 64-bit
Processor	12th Gen Intel Core i7-1260P, 12 cores, 16 logical processors
RAM	16 GB
Storage	1 TB SSD
Graphics	Intel® Iris® Xe Graphics
System Type	x64-based
BIOS Mode	UEFI

A smartphone and/or emulator was also used to test the mobile application during development.

3.13.2 Internet and Network Resources

Internet connectivity was used throughout the project for software development, documentation research, dependency installation, API access, map services, cloud-based development resources, and testing. Internet access was particularly required for the integration and testing of the system and for accessing online technical documentation.

3.13.3 Development Technologies and Frameworks

The main development technologies used in the implementation of DouaneNav included:

- **React Native with Expo** for the mobile application.
- **React.js** for the web dashboard.
- **Node.js and Express.js** for backend development and API services.
- **TypeScript** for strongly typed application development.
- **MongoDB** for database management.
- **Prisma** for database access and data modelling.
- **JWT and scrypt** for authentication and password security.
- **OpenStreetMap** for interactive map visualisation and destination display.
- **Axios** for communication between the application and backend services.

3.13.4 Development and Documentation Tools

Additional tools and resources were used for coding, system design, testing, documentation, and project management. These included a code editor, browser-based development and documentation resources, UML modelling tools, database development tools, and Microsoft Word for preparation of the project report.

These materials provided the technical environment required to design, implement, test, and document the DouaneNav system.

CHAPTER FOUR

PRESENTATION OF SYSTEM IMPLEMENTATION AND RESULTS

This chapter presents the results of the development and implementation of **DouaneNav: Destination Visualizer and Verification System for Cameroon Customs**. It focuses on the functionality of the developed system, the user interface, the interaction between the different system users, and the results obtained from testing each module. The chapter demonstrates how the various components work together to support customs officers in visualizing declared destinations, managing declarations, conducting inspections, generating reports, displaying prototype AI-assisted analysis results, and synchronizing data. Screenshots of the implemented interfaces are presented to illustrate the operation of each module and to demonstrate that the system meets the objectives defined in Chapter One.

The documentary analysis findings informed the functional requirements and design decisions implemented in the system. The following sections present the implementation and testing results of the developed application.

4.1 Presentation of Descriptive Statistics

In place of survey response frequencies, this section presents two kinds of descriptive data appropriate to a systems-development study: the implementation status of each of the eight planned modules, and the results of the automated tests run against the modules completed so far.

4.1.1 Development Status Overview

Table 6: Development status overview

Number	Module	Key Components	Status
1	Authentication	Login, Logout, JWT issuance, role-based access control, 3 seeded roles	Complete
2	Dashboard	Alerts, recent inspections, synchronisation status summary	Complete
3	Declarations	List, search, filters, details, documents	Complete
4	OpenStreetMap	Destination marker, geocoding, route preview	Complete

5	AI Module	Mock document analysis, destination extraction, inconsistency detection	Complete prototype only
6	Inspection Module	Create inspection, photos, comments, status, history	Complete
7	Synchronisation Module	Simulated sync, queue, sync history, CAMCIS reference	Complete
8	Reports	Daily reports, officer activity, inspection history, export	Complete

Development follows Clean Architecture, per the Software Architecture document: routes delegate to controllers, controllers delegate to services, and services alone interact with the MongoDB models, so each module can be built, tested, and verified independently before the next begins.

4.1.2 Module 1 (Authentication)

The authentication module provides secure access to the system through role-based login, session management and password recovery interfaces. The following screenshots demonstrate the authentication workflow implemented for both the mobile application and the web dashboard, including successful login, validation of invalid credentials, and password recovery pages.

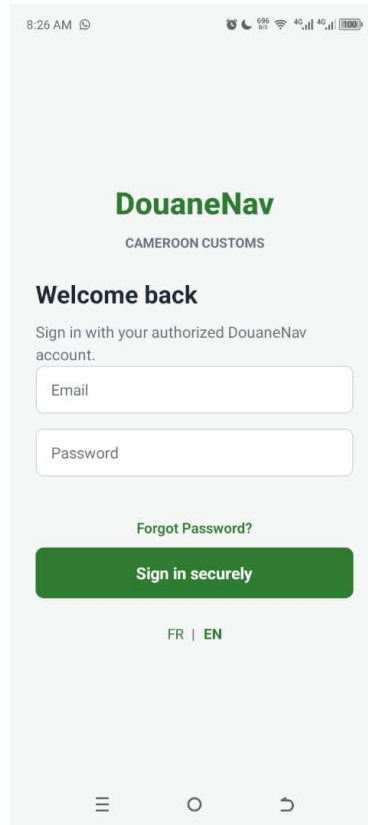


Figure 7: Login page

4.1.3 Automated Test Results

Automated tests were written and run against Module 1, covering both the underlying authentication logic (password hashing, JWT generation and verification, and the RBAC middleware's decision logic) and the HTTP layer (route wiring, input validation, and access control on protected endpoints). All tests passed.

Table 7: Test results

Test Category	Tests Run	Passed	Failed	Pass Rate
Password hashing (bcrypt)	3	3	0	100%
JWT token generation/verification	4	4	0	100%
RBAC middleware (role/permission guards)	5	5	0	100%
API routing	3	3	0	100%
Input validation	2	2	0	100%

Authorization (protected routes)	2	2	0	100%
TOTAL	19	19	0	100%

4.1.4 Modules 2–8

The following figures present the principal interfaces developed for the DouaneNav mobile application. Together, they illustrate the complete workflow of the proposed system, beginning with user dashboard, followed by declaration management, destination verification, inspection reporting, and synchronisation. Each screenshot demonstrates the corresponding functionality implemented during system development and provides visual evidence of the application's operation.

Module 2 — Dashboard

The dashboard provides an overview of the main operational information available in DouaneNav. It presents statistics, alerts, recent inspections, and synchronisation status, allowing authorised users to quickly monitor system activity.

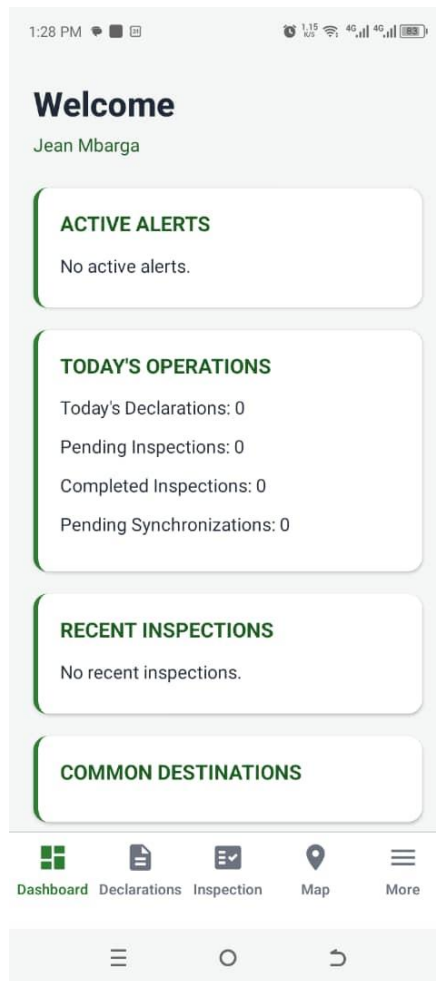


Figure 8: Dashboard

Module 3 — Declarations

The declaration module allows Customs Officers to retrieve and search declaration records using relevant criteria such as declaration number, truck registration plate, importer, destination, and date. The module also provides access to declaration details and associated documents required for verification.

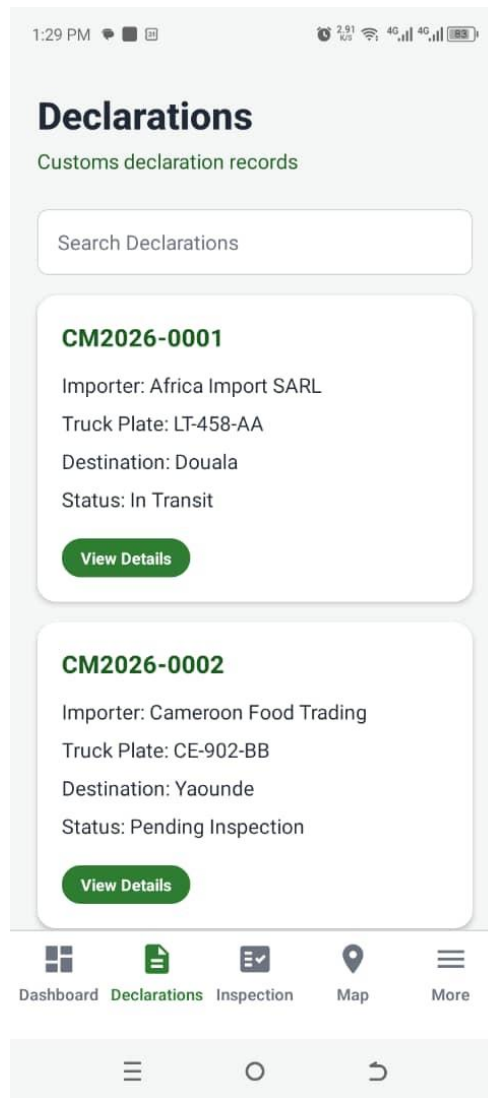


Figure 9: Declaration search results list

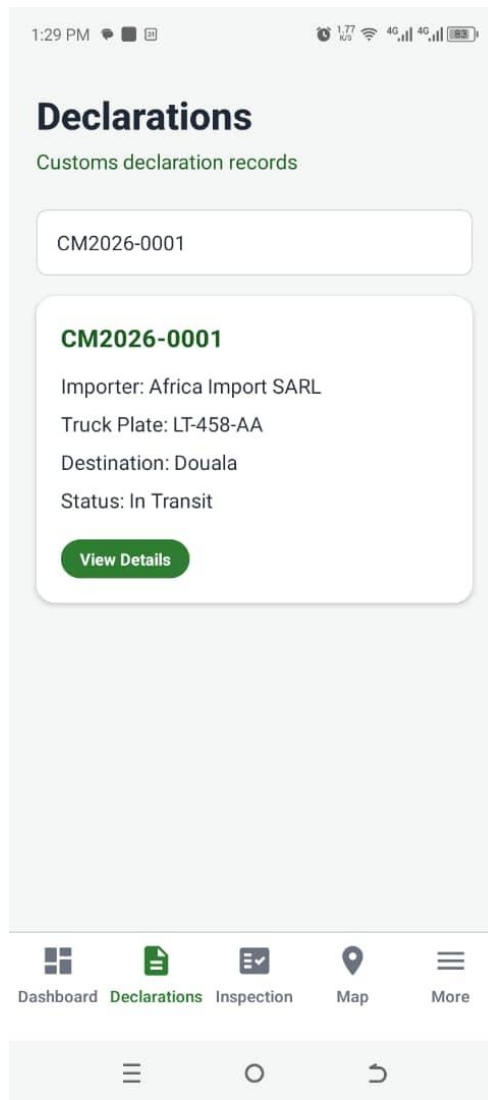


Figure 10: Search by declaration number

Module 4 — OpenStreetMap

The destination visualisation module displays the declared destination on an interactive OpenStreetMap-based map. Geocoding converts the destination into geographical coordinates, while the map and route preview help the officer understand the location and plan the inspection.

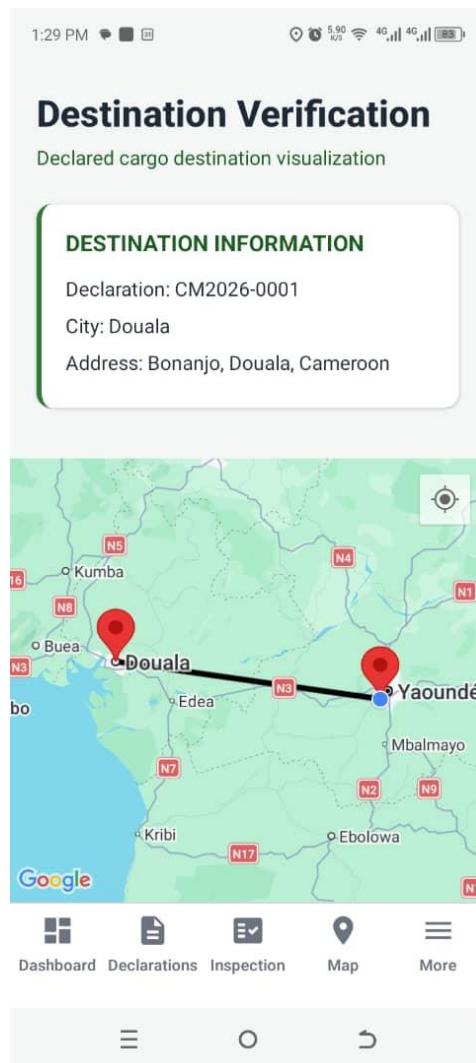


Figure 11: Full map screen with destination marker

Module 5 — AI Module (Mock)

The AI module demonstrates a prototype for document analysis within DouaneNav. The prototype extracts destination information from documents and identifies potential inconsistencies. This functionality remains a prototype and is presented as a basis for future enhancement.

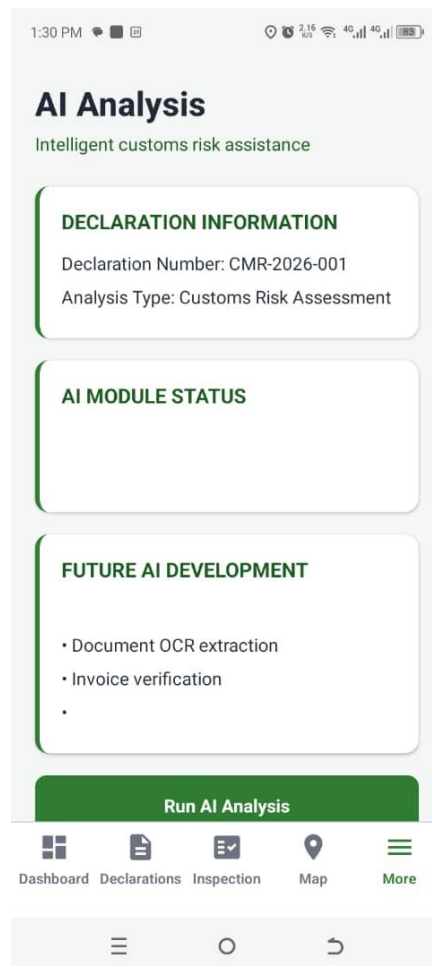
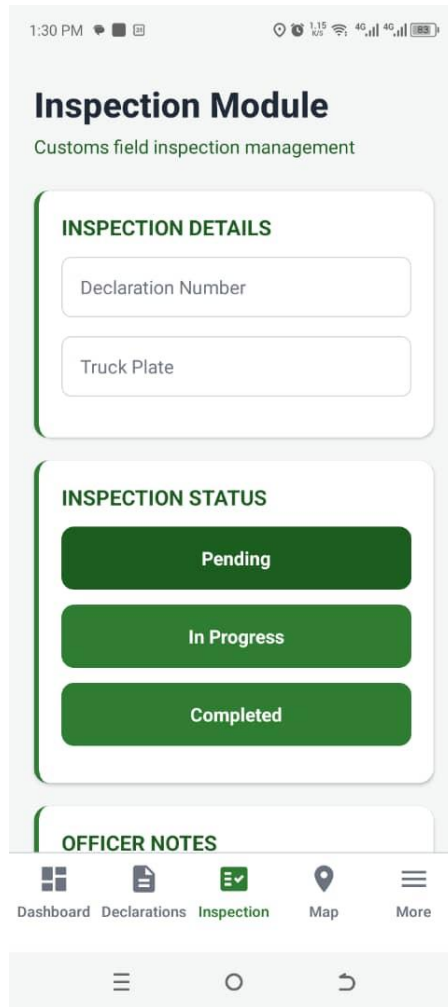


Figure 12: Mock AI analysis result

Module 6 — Inspection Module

The inspection module enables Customs Officers to record the results of a field inspection against a specific declaration. Officers can enter comments, record the inspection status, attach photographs, and consult the inspection history.



The screenshot displays the 'Inspection Module' interface on a mobile device. At the top, the status bar shows the time as 1:30 PM and various connectivity icons. The app header includes the title 'Inspection Module' and the subtitle 'Customs field inspection management'. The main content area is divided into three sections: 'INSPECTION DETAILS' with input fields for 'Declaration Number' and 'Truck Plate'; 'INSPECTION STATUS' with three green buttons labeled 'Pending', 'In Progress', and 'Completed'; and 'OFFICER NOTES' with a text input area. A bottom navigation bar features icons for 'Dashboard', 'Declarations', 'Inspection' (the active tab), 'Map', and 'More'. Below this is a standard Android-style navigation bar with back, home, and recent apps buttons.

Figure 13: Inspection form

Module 7 — Synchronisation Module

The synchronisation module manages completed inspection reports through a local queue and simulated synchronisation process. Inspection reports are retained locally when synchronisation cannot be completed and are made available for subsequent synchronisation attempts. The current implementation does not connect directly to the live CAMCIS platform because authorised integration was not available during the study. Instead, the synchronisation workflow and CAMCIS reference are simulated to demonstrate the intended integration process.

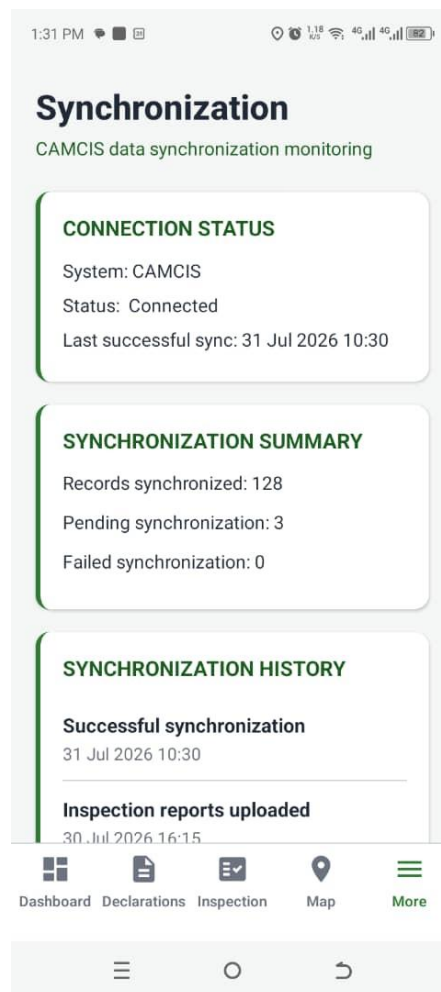


Figure 14: Synchronisation

Module 8 — Reports

The reports module provides structured information about completed inspections and officer activities. It supports daily inspection reports, inspection history, activity monitoring, and report export for further use and documentation.

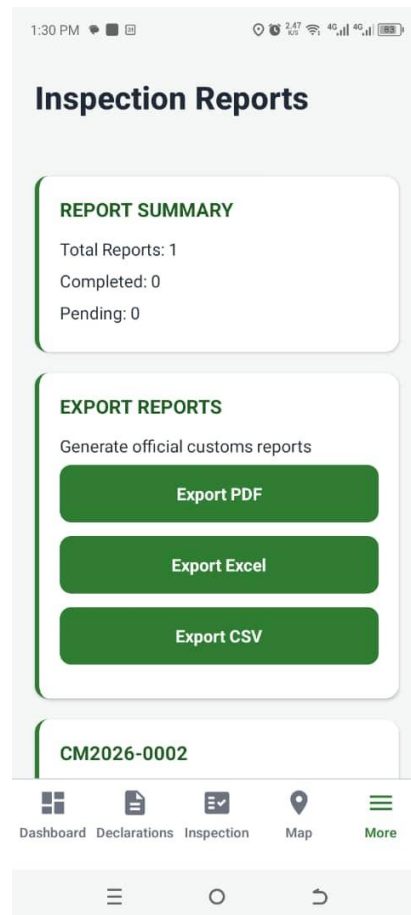


Figure 15: Report page

CHAPTER FIVE

DISCUSSION, CONCLUSION, RECOMMENDATIONS AND FUTURE WORKS

5.1 Discussion of Findings

The primary objective of this project was to develop a mobile-first decision support subsystem capable of assisting Cameroon Customs Officers in verifying declared cargo destinations through map visualisation, structured inspection reporting, and synchronisation with an administrative monitoring platform. The completed system achieved the implemented objectives defined for this study, while AI-based document analysis remains a prototype intended for future enhancement.

The authentication module provides secure role-based access control for Customs Officers, Supervisors, and Administrators. Declaration records can be retrieved and searched using relevant criteria, allowing officers to access important shipment information more efficiently. The destination visualisation module uses OpenStreetMap to display declared destinations on an interactive map, providing a more intuitive representation of location information than text-based records alone.

The inspection module enables officers to create structured inspection reports, record inspection outcomes and comments, and attach photographic evidence. This provides a centralised digital record of inspection activities associated with individual declarations.

The synchronisation module manages completed inspection reports through a local queue and synchronisation process. This allows reports to remain available when connectivity is temporarily unavailable and supports their transmission when the connection becomes available. The reporting module further provides structured information on inspections and officer activities for administrative monitoring.

The web dashboard provides administrative and supervisory functions, including system statistics, inspection monitoring, synchronisation status, and user management according to the assigned role.

The Artificial Intelligence module was implemented only as a mock prototype. It demonstrates the intended direction for document analysis and destination information extraction but does

not constitute a production-ready artificial intelligence system. Full AI-powered document analysis is therefore considered a future enhancement.

Overall, the findings demonstrate that the developed prototype satisfies the functional requirements established during the analysis and design phases. The system provides a practical foundation for improving access to declaration information, destination visualisation, field inspection recording, and administrative monitoring.

5.2 Discussion of the Project Plan

The project was executed according to the implementation plan established during the system design phase. Development was organised into independent but interconnected modules, allowing individual components to be implemented, tested, and refined before integration into the complete system.

The implementation sequence consisted of:

- Authentication and Role-Based Access Control
- Dashboard Development
- Declaration Management
- Destination Visualisation using OpenStreetMap
- AI Module Prototype
- Inspection Management
- Synchronisation Services
- Reporting Module
- Administrative Web Dashboard

The modular development approach simplified debugging, improved maintainability, and allowed individual components to be tested independently before system integration.

Several refinements were introduced during development, including interface improvements, English and French localisation, improved synchronisation feedback, and enhanced inspection report presentation. These modifications improved the usability and consistency of the final prototype while remaining within the defined project scope.

The project was nevertheless subject to several constraints. Direct integration with the live CAMCIS platform was not available during the study, and development therefore relied on simulated declaration data. In addition, the project was developed by a single student within a limited academic schedule. These constraints were considered during system design and are reflected in the future development requirements presented later in this chapter.

5.3 Project Closeout Decision

The project has successfully achieved its implemented objectives and is considered technically feasible as an academic prototype for future deployment within the operational environment of Cameroon Customs, subject to institutional approval, security assessment, and authorised integration with CAMCIS.

The developed system demonstrates that customs destination verification, structured inspection reporting, synchronisation management, and administrative monitoring can be implemented using a mobile-first architecture that complements rather than replaces the existing customs declaration system.

However, the current system should not be considered a production deployment. Live CAMCIS integration, real-world operational testing, and several advanced features remain necessary before institutional deployment.

Consequently, the project is considered successfully completed within the defined academic scope, while providing a foundation for further development and eventual operational evaluation.

5.4 Conclusion

This project successfully designed and implemented DouaneNav, a mobile-first decision support subsystem intended to complement CAMCIS and support customs field inspection activities.

The implemented system provides secure authentication and role-based access control, declaration retrieval and search, destination visualisation using OpenStreetMap, structured inspection reporting, synchronisation management, reporting functions, and a complementary web dashboard for administrative and supervisory monitoring.

The system demonstrates how mobile technologies and geographic visualisation can be combined with structured digital inspection reporting to provide customs officers with more accessible information during field activities. Rather than replacing CAMCIS or the judgement of Customs Officers, DouaneNav is designed to complement the existing customs information environment and support operational decision-making.

The project also demonstrates the importance of designing systems that account for connectivity constraints. The local storage and synchronisation approach provides a basis for maintaining inspection information when continuous connectivity cannot be guaranteed.

Although direct integration with live CAMCIS data and advanced AI capabilities were outside the scope of the implemented prototype, the modular architecture provides a foundation for these and other future enhancements. The study therefore demonstrates the technical feasibility of developing a complementary mobile-first solution for customs destination verification and inspection management.

5.5 Policy Implication

The development of DouaneNav demonstrates the potential value of complementary mobile technologies within public-sector customs operations.

For Cameroon Customs, the adoption of complementary digital tools could support improved access to operational information, more structured inspection documentation, and better monitoring of field activities. Such systems should, however, be integrated within existing institutional processes rather than developed as replacements for established customs platforms such as CAMCIS.

Any future institutional deployment should therefore consider data security, user access control, interoperability, connectivity requirements, system maintenance, and formal authorisation for access to customs information.

The project also illustrates how geographic information technologies and mobile applications can contribute to broader public-sector digital transformation by making operational information more accessible to personnel working outside fixed office environments.

5.6 Recommendations

Based on the findings of the study, the following recommendations are proposed:

- **Cameroon Customs** should consider evaluating complementary mobile solutions for field inspection activities where access to declaration information and structured reporting is required.
- **Relevant technical authorities** should establish appropriate interoperability and security requirements before connecting complementary applications to CAMCIS or other official customs systems.
- **Institutional stakeholders** should conduct controlled pilot testing in selected customs offices before any wider deployment, in order to evaluate usability, reliability, security, and operational suitability under real working conditions.
- **Customs personnel** should receive appropriate training before the deployment of any new mobile inspection system to ensure that the technology supports existing procedures rather than creating additional operational complexity.
- **System administrators** should maintain appropriate access controls, audit mechanisms, data protection measures, and system maintenance procedures to protect sensitive customs information.

5.7 Future Works

Several features and improvements remain possible beyond the scope of the present study.

5.7.1 Live CAMCIS Integration

Future development should establish an authorised API or other secure integration mechanism with the live CAMCIS platform. This would replace the simulated declaration dataset used during development and allow DouaneNav to retrieve actual authorised declaration information.

5.7.2 AI-Powered Document Analysis

The current AI component is only a mock prototype. Future work could implement a production-ready document analysis system capable of extracting information from customs documents, comparing document contents, identifying inconsistencies, and supporting risk assessment.

5.7.3 Advanced Routing and Navigation

Future versions could integrate a suitable routing and navigation service to provide real road routes between an officer's location and the declared destination. Any such integration would depend on institutional requirements, technical availability, and licensing considerations.

5.7.4 Advanced Report Export

Future versions could extend the reporting functionality with additional export formats and automated generation of administrative reports, including PDF, Excel, or CSV outputs where required by the institution.

5.7.5 Notifications

Push notifications could be introduced to alert Supervisors or Administrators when new inspection reports are synchronised, when synchronisation failures require attention, or when other defined operational events occur.

5.7.6 Offline Mapping

Offline map caching could be implemented to improve destination visualisation in areas where internet connectivity is limited or temporarily unavailable.

5.7.7 Advanced Analytics

Future versions could introduce more advanced statistical dashboards and operational analytics to support the identification of inspection trends, officer activity patterns, synchronisation performance, and other relevant indicators.

5.7.8 Real-World Evaluation

A controlled pilot deployment within selected Cameroon Customs offices would provide an opportunity to evaluate the system under real operational conditions. Such evaluation could measure usability, response time, reliability, connectivity performance, and the practical value of destination visualisation and structured digital inspection reporting.

5.7.9 Google Maps SDK Integration

Although the current implementation uses OpenStreetMap for destination visualisation, future versions may integrate the **Google Maps SDK/Google Maps Platform**, subject to the

applicable API, billing, licensing, and institutional requirements. This could provide additional mapping and routing capabilities where required by the operational environment.

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APPENDICES

APPENDIX A: SOURCE CODE SCREENSHOTS

This appendix presents selected screenshots of the source code developed for the DouaneNav system. The screenshots are organised according to the main functional features implemented in the system and provide evidence of the development work described in Chapter Four.

A.1 Authentication Module

Figure A1: Login Form Source Code



```
mobile-app-expo > src > screens > LoginScreens > LoginScreen > handleLogin
38 export default function LoginScreen({
39   const handleLogin = async ()=>{
40     |email ||
41     |password
42   }{
43
44     Alert.alert(
45       "Missing information",
46       "Please enter your email and password."
47     );
48
49     return;
50   }
51
52   try{
53
54     await login(
55       email.trim(), password
56     );
57
58     Alert.alert([
59
60       "Success", "Login successful"
61     ]);
62
63   }
64
65   catch(error){
66
67     Alert.alert(
68       "Login failed",
69       error.response?.data?.error
70       ||
71       error.message
72       ||
73       "Unable to login."
74     );
75
76   }
77
78 }
```

A.2 Dashboard Module

Figure A2: Dashboard Source Code

```
export default function DashboardScreen({
  const loadDashboard = async()=>{
  });

  catch(error){
    console.log(
      "Dashboard error:",
      error.message
    );

    setDashboard({
      counts:{
        todayDeclarations:0,
        pendingInspections:0,
        completedInspections:0,
        pendingSynchronizations:0
      },
      alerts:[],
      recentInspections:[],
      destinationStats:[],
      inspectionStatistics:{
        total:0,
        completed:0,
        pending:0
      }
    });
  }
});
```

APPENDIX B: ADDITIONAL SYSTEM SCREENSHOTS

Additional screenshots of the implemented system may be included in this appendix where necessary to provide further evidence of functionality not fully shown in Chapter Four.

Web Dashboard Screenshots

Administrator Dashboard

DouaneNav

FR EN

Dashboard

Declarations

Reports

Inspection Monitoring

Synchronization

Users

Configuration

Audit Logs

Logout

CAMEROON CUSTOMS

Operational Dashboard

Monitoring declarations, inspections and synchronization.

Today's Declarations

0

Pending Inspections

0

Completed Inspections

0

Pending Synchronizations

0

Recent Inspections

Declaration	Officer	Destination	Result	Status	Date
				Completed	
	Estelle Fongang			Pending	

Synchronization Status

Successful %

Pending

Failed

Last synchronization

Active Alerts

Declaration Statistics

User Management Interface

DouaneNav

FR EN

Dashboard

Declarations

Reports

Inspection Monitoring

Synchronization

Users

Configuration

Audit Logs

Logout

CAMEROON CUSTOMS

User Administration

Manage users, roles and permissions.

User Management

CustomsTrack AI authorized users.

Name	Email	Role	Status
Amina Ndzi	admin@douanenav.cm		active
Jean Mbarga	officer@douanenav.cm		active
Estelle Fongang	brigade@douanenav.cm		active
Paul Tchana	supervisor@douanenav.cm		active

C

APPENDIX C: QUESTIONNAIRE

No questionnaire was used in this study. The research relied primarily on documentary sources and system development activities; therefore, no questionnaire is included in this appendix.